



**Verified Carbon
Standard**

SCATEC RMI HYBRID CLUSTER

Scatec

Anthesis 

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The Scatec RMI Hybrid Cluster (the "Project") is a single site with multiple (3) equivalent project activities, dispatchable power facility, comprising of solar photo-voltaic power feeding into a battery energy storage system (BESS) technology prior to feeding into the national grid at the Rooidam switching station. The Project is located approximately 20 km North-East of the town of Kenhardt in the Northern Cape province of South Africa and is located near Eskom's Nieuwehoop Substation. The Project has a Contracted Capacity of 150 MW at the Delivery Point and comprises an area of around 880ha. The solar PV portion is split over three project activities located adjacent to one another in close proximity to the central BESS, Rooidam switching station and Operating infrastructure and forms a combined total installed capacity of approximately 540 MW peak. The BESS has a combined energy capacity of 1140 MWh, and 225 MW dispatch capacity, at the Delivery Point.

The Project achieved its Commercial Operation Date (COD) with the signature of the Notice of Commencement of the Facility between the facility operating entity and the national power utility formally commencing the Power Purchase Agreement [1]. This was achieved for the first project activity (cluster) of the project on 18 November 2023. This is the start date of the project and its associated GHG reduction activity. From this date, the plant operated, with remaining two project activities (clusters) going live with the achievement of COD on 25 November 2023 and 8 December 2023 respectively. No other changes from the project implementation outlined in this document were experienced in this initial verification period. No additional project activities (Cluster) will be added to the project. Financial close for the project was reached in June 2022 with construction commencing in July 2022 [2]. All Operations and Maintenance of the project is conducted by a subsidiary Scatec company at the start date.

Prior to implementing this project, electricity was supplied by the current fleet of power plants owned and operated by Eskom, the majority of which are fossil fuel based. The Renewable Energy Independent Power Producer Procurement Programme (REIPPPP) is an initiative of the South African government aimed at increasing electricity capacity through private sector investment in solar photovoltaic and concentrated solar, onshore wind power, small hydro, landfill gas, biomass, and biogas [3]. Since coal is an emissions-intensive energy carrier, South Africa's economy is very emissions-intensive, especially when compared to other countries at a similar stage of development as measured by the Human Development Index [4]. South Africa's energy sector is by far the largest contributor to the country's emissions, contributing 94.7% to the country's predominant GHG, which is carbon dioxide (CO₂) [5].

On average the project activity is expected to reduce annual GHG emissions equivalent to circa 830,000 tCO₂e and 8,300,000 tCO₂e over the crediting period of 10 years.

The total GHG emission reductions generated in the monitoring period from 18-November-2023 to 28-February-2025, is 1,045,198 tCO₂e

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
<i>Validation</i>	18-November-2023 – 17-November-2033	VCS	Carbon Check (India) Private Ltd.	Ten years (10) non renewable
<i>Verification</i>	18-November-2023 – 28-February-2025	VCS	Carbon Check (India) Private Ltd.	1.25 years

1.3 Sectoral Scope and Project Type

Sectoral scope¹	Scope: 01 - Energy (renewable / non-renewable)
Project activity type	Renewable Energy Project

1.4 Project Eligibility

1.4.1 General eligibility

The project is eligible under the scope of the VCS Program since the project activity meets the criteria listed in paragraph 2.1.1 of the VCS Standard, Version 4.7. Table 1 below describes the scope of the VCS program and the project's applicability, and Table 2 further elaborates on the project's eligibility .

Table 1: Scope of the VCS program

Elements	Justification of project applicability
1) The seven Kyoto Protocol greenhouse gases.	The project aims at reducing greenhouse gas (GHG) emissions by replacing fossil fuel energy supplied to

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

	the South African national grid, with clean renewable electricity.
2) Ozone-depleting substances (ODS).	Not applicable
3) Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.	Project activity applies VCS-approved CDM methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0) ² which has been approved for use under the VCS Program.
4) Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).	Project activity applies VCS approved CDM methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0) ³ .
5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the Jurisdictional and Nested REDD+ (JNR) Requirements.	N/A

Table 2: Excluded Project Activities under the VCS v 4.7 Table 1

Activity	Applicability
Grid-connected electricity generation activities using hydroelectric power plants. The exclusion does not apply to ocean energy (e.g., wave, tidal, salinity gradient, and ocean thermal energy conversion). For hydro projects, large scale means a maximum capacity of greater than 15MW, where maximum capacity is the installed/rated capacity or authorized capacity (as determined in the activity approval from the project regulator, government, or similar entity), whichever is lower. Grid-connected means >50% of total generation is exported to a national or regional grid.	Not applicable to this project which is not a hydroelectric power plant.
Grid-connected electricity generation activities using wind, geothermal, or solar photovoltaic (PV) power plants. The exclusion does not apply to concentrated solar thermal-to-electricity, floating solar PV, or energy storage systems (e.g., batteries). Grid-connected	This is a grid-connected electricity generation activity using solar photovoltaic (PV) to supply the BESS

² <https://cdm.unfccc.int/methodologies/DB/XB1TX7TAZ6SLWM9B7BC67THHVD16JV>
³ <https://cdm.unfccc.int/methodologies/DB/XB1TX7TAZ6SLWM9B7BC67THHVD16JV>

means >50% of total generation is exported to a national or regional grid. See the VCS Program Definitions for the full definition.	technology. Hence the project is eligible and not excluded.
Activities recovering waste heat for combined cycle electricity generation, or to heat/cool via cogeneration or trigeneration. The exclusion does not apply to waste gas recovery or electricity generation using waste heat recovery outside of combined cycle applications (e.g., organic Rankine cycles).	Not applicable to this project which is not a waste heat recovery project.
Activities generating electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass, or forest residue biomass. The exclusion does not apply to gasification, pyrolysis, combusting biofuels, biogas, fractions of renewable biomass in refuse-derived fuels, agro/forest biomass residues in waste streams that are sent to landfills, CO ₂ capture and storage from renewable biomass combustion, or thermal efficiency improvements (e.g., cook stoves).	Not applicable to this project which does combust biomass.
Activities generating electricity and/or thermal energy using fossil fuels, and activities that involve switching from a higher to a lower carbon content fossil fuel. The exclusion does not apply to the use of captured flare and/or vent gas or waste containing previously used petroleum products (e.g., used plastics, oils, lubricants).	Not applicable to this project which does not combust fossil fuel.
Activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs). Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO ₂ e per year.	Not applicable to this project which does not replace electric lighting.
Activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers. Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO ₂ e per year.	Not applicable to this project which does not install or replace electricity transmission lines or transformers.
Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions	Not applicable to this project which does not

	reduce HFC-23 emissions.
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Pipeline listing and validation deadlines: The project activity was listed on the VERRA website in April 2025, to initiate the validation process. The opening meeting with the VVB took place once the project activity had been listed and showed the status of 'Under Validation' in the VCS Program Registry. Furthermore the VVB did not complete validation until the 30-day public comment period had ended. As per para 3.8.1 of the VCS standard V4.7, non-AFOLU projects shall complete validation within two years of the project start date and this condition will be fulfilled by this project activity as the start date of the project activity is 18 November 2023 which is less than two years prior to the completion of validation.

The project activity applies CDM methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0), which has been approved by Verra for use under the VCS Programme for project activities seeking registration with them. ACM0002 is a large-scale methodology and has no capacity limit, and the Scatec RMI Hybrid Cluster is a multiple project activity project in one location and is not part of a grouped project, a larger project or activity.

1.4.2 AFOLU project eligibility

This section is not applicable, as the project is not an AFOLU project.

1.4.3 Transfer project eligibility

This section is not applicable, as the project is not being transferred from another GHG Program nor a CPA seeking registration under the VCS Program.

1.5 Project Design

- ☐ Single location or installation
- ☒ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

Grouped Project Design

This is not a Grouped project. Hence, this section is not applicable.

1.6 Project Proponent

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1.7 Other Entities Involved in the Project

There are no other entities involved.

1.8 Ownership

The three Scatec Project Proponents. have the legal right to own the Scatec RMI Hybrid Cluster facilities, which are verified through the following documents:

- Eskom's Power Purchase Agreements dated June 2022 which includes all details of the physical locations and property boundaries.
- The Scatec Project Proponents have an agreement with subsidiaries of the parent company (Scatec ASA) for functions such as Operations & Maintenance, Development Work, EPC, Management Services and other shared services, including the Global Operations Centre (GOC).
- NERSA Generation Licences dated July 2022. [xx]
- The Operator has a legal agreement and right to use the land with land lease agreements for the duration of the project lifetime, which can be verified through the following documents provided by the project proponent:
 - Lease agreements exist for all clusters and operations. The lease period is for a minimum of 20 years from COD. The lease covers the footprint of the land – Remaining Extent of the farm Onder Rugzeer Number 168; Division Kenhardt; Northern Cape [xx]
- Anthesis and the Scatec PPs have a legal agreement in place for the joint ownership of a registered carbon project. [xx]

1.9 Project Start Date

Project start date	18-November-2023
Justification	As per §3.8 of the VCS Standard, the start date “is the date on which the project began generating GHG emission reductions.” The project activity started generating GHG Emission removals on the Commercial Operations Date (COD) which was signed and confirmed as being 18 November 2023 [6]

1.10 Project Crediting Period

Crediting period	☐ Seven years, twice renewable ☒ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	18-November-2023 to 17-November-2033

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals (ERRs) of the project:

- ☐ < 300,000 tCO₂e/year (project)
☒ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
18-November-2023 to 31-December-2023	83 913
01-January-2024 to 31-December-2024	824 868
01-January-2025 to 31-December-2025	830 000

01-January-2026 to 31-December-2026	830 000
01-January-2027 to 31-December-2027	830 000
01-January-2028 to 31-December-2028	830 000
01-January-2029 to 31-December-2029	830 000
01-January-2030 to 31-December-2030	830 000
01-January-2031 to 31-December-2031	830 000
01-January-2032 to 31-December-2032	830 000
01-January-2033 to 17-November-2033	605 000
Total estimated ERRs during the first or fixed crediting period	8 300 000
Total number of years	10
Average annual ERRs	830 000

1.12 Description of the Project Activity

The Scatec RMI Hybrid Cluster project (the "Project") is a multiple project activity on a single site, dispatchable facility, comprising of solar photo-voltaic renewable energy generating technology feeding into a battery energy storage system (BESS). The solar photo-voltaic components form a cluster around the central BESS and transformer switching station that feeds the national grid at the Rooidam switching station on site. The Project is located approximately 20 km North-East of the town of Kenhardt in the direction of Groblershoop in the !Kheis Local Municipality of the Northern Cape province of South Africa and is located near Eskom's Nieuwehoop Substation. The Project has a Contracted Capacity of 150 MW (AC) at the Delivery Point. The solar portion is spread over 6 clusters located adjacent to one another and forming a combined total installed capacity of an approximately 540 MW peak (DC). The BESS has a combined energy capacity of 225 MW dispatchable capacity, at the Delivery Point.

A project owned 132/33kV switching step-up substation (Rooidam switching station) is located within the project footprint with a 85 MVA 132/33kV Step-up transformer and transformer bay. The isolators on the HV Site of the IPP Transformer are the location of the Main Metering Installation, the Point of Connection (POC) and where energy will be exported from the Project to the national grid at the Delivery Point. The interconnection works include a 132 kV switching station, located adjacent to the IPP Substation, connected to the 132 kV network.

Project or Facility Details

Project Name	Scatec RMI Hybrid Cluster
Contracted Capacity [MW]	150 MW (3 x 50MW equivalent project activities)
Design Auxiliary Consumption of the Facility [MW]	Solar, 0.72 MW. Battery, 2.25 MW
No. of PV clusters	6
No. of PV panels	990 864
PV Module type	JAM72D30-545 (540, 545, 550 Wp)
Tracker system	PVH Axone Duo (single-axis) (55° East to 55° West) See Figure 1 below
Type of inverter manufacturer	Huawei SUN2000 200KTL
PV Inverter units	2232
PCS Inverter type	BYD PSC9567-3150
PCS Inverter units	180
Number of Transformers:	MV: 30 - 37X Hv:1xTBD
Planned voltage connection level:	132 kV
Emergency auxiliary genset	13 gensets on site to power control circuit if electrical system is down 13 gensets on site
Substation and transformer	Actom (Rooidam Substation)
Connection point (i.e. the name of the substation or distribution line onto which the Facility is intended to connect):	Rooidam Switching Station – Eskom POC on the HV Side of the IPP owned Step - up Transformer at the IPP project site.

	Rooidam will have a 132 kV network that will then feed into the Eskom Nieuwehoop station nearby
Battery Energy Storage System (BESS)	BYD Cube Pro
Battery units	456
PV Cleaning	PV modules will be cleaned robotically with mechanical brushes on a rolling schedule

This is a Greenfield project activity generating renewable electricity from the solar photovoltaic component feeding the BESS technology and supplying it to the national grid. In the baseline scenario the equivalent of electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the TOOL 07 “Tool to calculate the emission factor for an electricity system”, version 7.0. There is no technology transfer occurring in the project activity. There was no project related activity on the site of the project owner prior to the implementation of this project. Hence pre-project scenario and baseline scenario are the same.

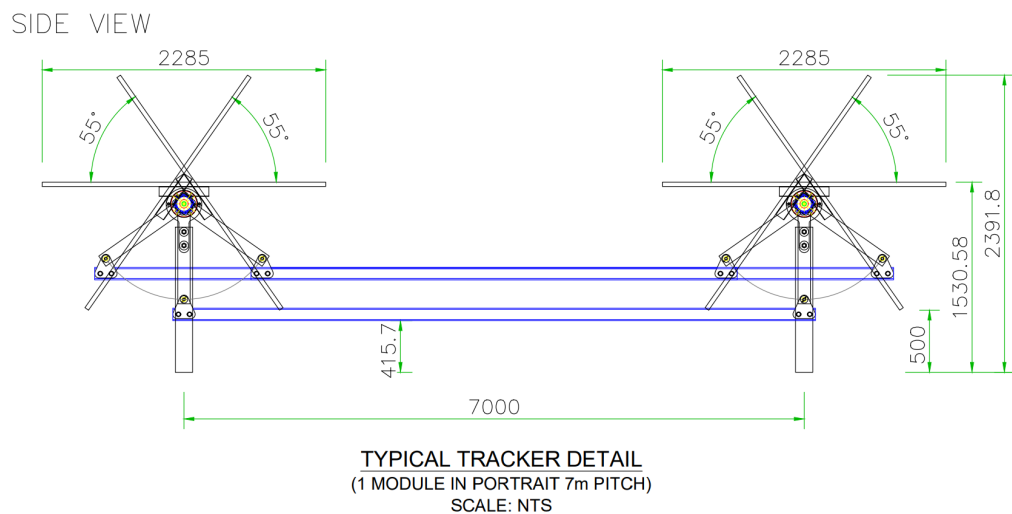


Figure 1: Typical PV Tracker system configuration

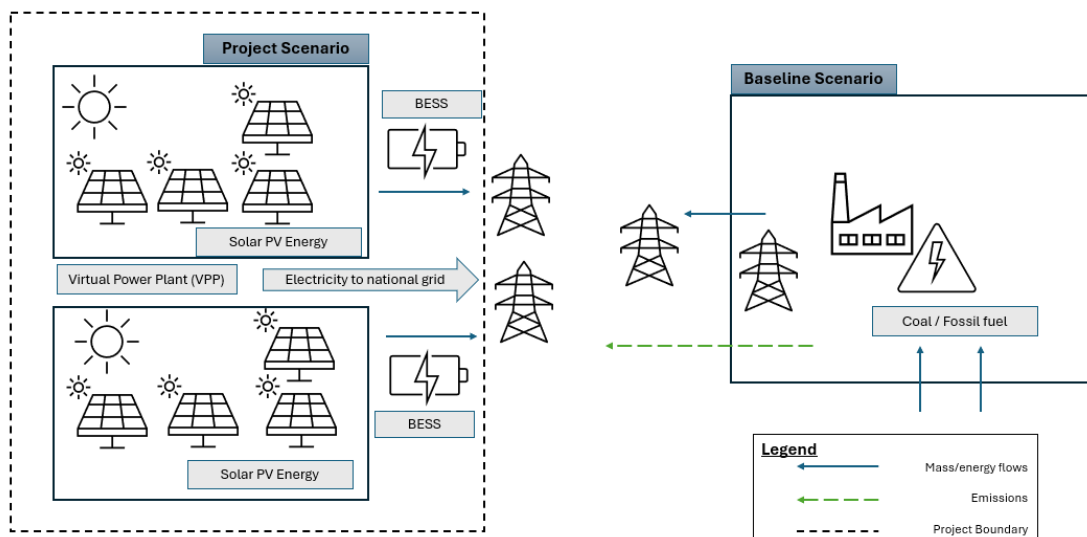


Figure 2: Schematic of project activities

1.13 Project Location

Project physical location and geographic coordinates	
Physical address	Remaining Extent of the farm Onder Rugzeer Number 168; Division Kenhardt; Northern Cape
Latitude	29° 12' 19.55"S
Longitude	21° 18' 5.54"E

A KML file has been provided on the VERRA Registry that includes the area of the multiple project activities around this point.

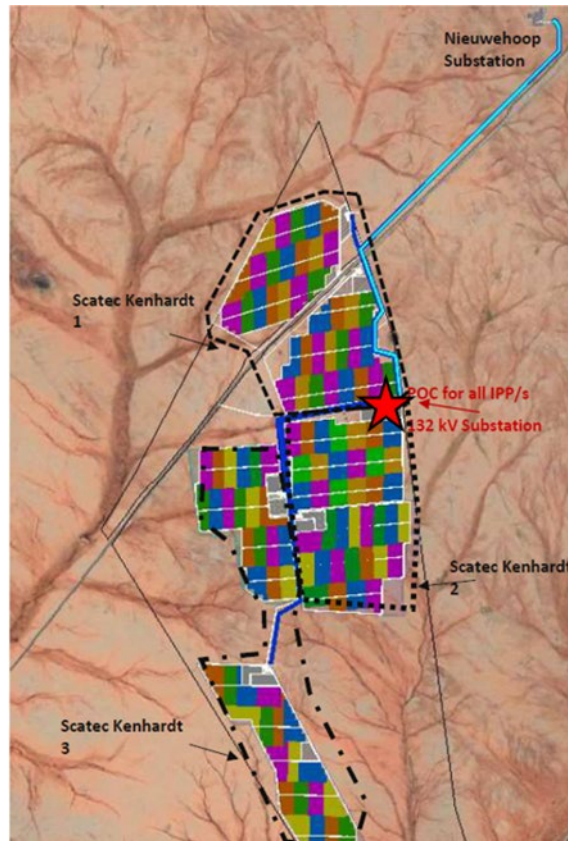


Figure 3: Physical location and layout of the multiple project activities on site

1.14 Conditions Prior to Project Initiation

Ecosystem type

The project site is located in the Nama Karoo biome which is characterized by semi-arid climatic conditions and vegetation that is dominated by shrubland. The ecosystem supports burrowing faunal species such as the *Suricata suricata*, and larger mammals such as the *Antidorcas marsupalis* which are not reliant upon the study area but are likely to forage over extensive ranges beyond the study area [7].

Current and historical land-use

The current land-use of the project site is a railway line known as the Sishen-Saldanha railway line which runs through the farm to the south of the project site with its associated infrastructure. The project site is located within a sheep farming agricultural region with no history of land cultivation on the farm. Due to the arid conditions and soil limitations, the site is

not suitable for any agricultural land use other than low-intensity grazing. Present and prior environmental conditions of the project area:

Climate

The average annual temperature of the Kenhardt area is 19.6°C, with highs reaching 33°C in summer, and an annual average low temperature of 11°C in winter. The area receives very low annual rainfall levels of 183 mm that occur mainly during the autumn season. Since temperatures are still relatively high in autumn, the associated evaporation rates are high, leading to a semi-arid climate.

Topography

The region wherein the project site is located can be described as a generally level portion of land with a low gradient and low open hills. The low vegetation and shrubland produce a mottled background which is effective at making development types such as power lines and pylons blend in with the background. [7]

Hydrology

The project site lies across a low ridge that bisects the area into two watersheds. The area has many ephemeral drainage lines which feed a major water course: Wolfkopseloop. This water course feeds into the Hartebees River which flows into the Orange River to the north. The groundwater quality is described as poor due to the low groundwater recharge conditions as a result of low rainfall and high evaporation rates.

Soils

The soil type comprises of predominantly shallow, red sands to loamy sands that fall into the arid Silicic, Calcic and Lithic soil groups. The project site has low to moderate water erosion hazard, mainly due to the low slope, but it is susceptible to wind erosion due to the sandy texture of the soil.

Vegetation

The project site is classified as a Bushmanland Arid Grassland veld type, and is dominated by *Rhigozum* shrubs and small trees, as well as *Aristida* grasses. 99% of the vegetation in the site remains intact, therefore the conservation status of the vegetation type is classified as Least Threatened, with only two species of provincial conservation significance, namely the *Aloe dichotoma* and the *Aloe claviflora*.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

Compliance with provincial, national and international legislation was considered during the planning and implementation of this project. The following legislative aspects were considered:

Constitution of the Republic of South Africa (Act 108 of 1996)	The Bill of Rights, in the Constitution of South Africa (No. 108 of 1996), gives everyone the right to a non-threatening environment and requires that reasonable measures are applied to protect the environment. This protection encompasses preventing pollution and promoting conservation and environmentally sustainable development. These principles are embraced and given further expression in NEMA.
National Development Plan, Agenda 2030	The NDP envisages that, by 2030, South Africa will have an energy sector that provides reliable and efficient energy service at competitive rates, that is socially equitable through expanded access to energy at affordable tariffs and that is environmentally sustainable through reduced pollution.
Environmental Conservation Act (No. 73 of 1989)	The ECA provides for the effective protection and controlled utilisation of the environment and for matters incidental thereto.
National Environmental Management Act, (No. 107 of 1998)	NEMA requires adherence to the principles of Integrated Environmental Management (IEM) in order to ensure sustainable development, which, in turn, aims to ensure that environmental consequences of development proposals be understood and adequately considered during all stages of the project cycle and that negative aspects be resolved or mitigated, and positive aspects enhanced.
National Water Act (No. 36 of 1998)	Provides the legal framework for the effective and sustainable management of our water resources.
Conservation of Agricultural Resources Act (No. 43 of 1983)	CARA provides for control over the utilization of the natural agricultural resources in order to promote the conservation of the soil, water sources and vegetation and the combating of

	weeds and invader plants; and for matters connected therewith.
Occupational Health and Safety Act (No. 85 of 1993)	OHSA requires the employer to bring about and maintain, as far as reasonably practicable, a work environment that is safe and without risk to the health of the workers. This means that the employer must ensure that the workplace is free of hazardous substances, such as benzene, chlorine and micro-organisms, articles, equipment, processes, etc. that may cause injury, damage or disease. Where this is not possible, the employer must inform workers of these dangers, how they may be prevented, and how to work safely, and provide other protective measures for a safe workplace.
The Basic Conditions of Employment Act (No. 75 of 1997)	The BCEA intends to give effect to the right to fair labour practices referred to in section 23(1) of the Constitution by establishing and making provision for the regulation of basic conditions of employment; and thereby to comply with the obligations of the Republic as a member state of the International Labour Organisation; and to provide for matters connected therewith.
Electricity Regulation Act (Act 4 of 2006) and the Electricity Regulation on New Generation Capacity R399 of 4 May 2011, as amended by R1366 of 04 November 2016	The ERA establishes a national regulatory framework for the electricity supply industry. It designates the National Energy Regulator of South Africa (NERSA) as the custodian and enforcer of this framework. The act covers licensing, registration and regulation of electricity generation, transmission, distribution, trading, import and export. The Regulations provide for the procurement of new generation capacity derived by organs of state, from renewable energy sources and co-generation; base load, mid-merit load and

	peak load new generation capacity; and cross border projects.
Integrated Resource Plan (IRP2019), and the draft IRP 2023	The IRP is the electricity infrastructure development plan, of the Department of Energy, based on the least-cost electricity supply and demand balance, taking into account security of supply and the environment (minimize negative emissions and water usage).
Carbon Tax Act 15 of 2019 and supporting regulations	Provide for VCU's to be used to offset carbon tax liability.

Table 3: Relevant Legislation

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

If yes, provide required evidence of no double issuance as outlined by the VCS Standard.

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes

☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes

☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes

☒ No

1.18 Sustainable Development Contributions

1.18.1 Sustainable Development Contributions Activity Description

The sustainability objective of the project lies in its capacity to reduce greenhouse gas emissions. In alignment over the seven-year crediting period, renewable twice, the project anticipates an average annual emission reduction of 830,000 tons of CO₂ equivalent, totalling approximately 7,891,789 tons of CO₂ equivalent over the project lifetime. This substantial reduction aligns with Sustainable Development Goal (SDG) 13 (Climate Action), specifically addressing the urgent need to combat climate change and its impacts. Monitoring will be undertaken as per the plan in this Project Description, which involves the monitoring of generated renewable energy in MWh/year. This further aligns with South African SDG 7.2 which aims to increase the renewable energy within the country. The project displaces approximately 860,000 MWh per annum of emission-intensive grid electricity through the generation of low-emission renewable electricity.

Furthermore, aligned with SDG 8.5, the project will promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all. The operational phase will create 234 permanent employment opportunities for the lifetime of the project. The short-term employment opportunities accounted for in the construction phase of the project play a

pivotal role in the job seekers' employability as they will acquire skills that will enhance their prospects for future job opportunities.

1.18.2 Sustainable Development Contributions Activity Monitoring

Table 4: Sustainable Development Contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
1)	7.2	7.2.1A1 (South African SDG Indicators) – Amount of renewable energy at annual operating capacity	Implemented activities will increase the amount of renewable energy in South Africa	Displaces approximately 830,000 MWh per annum of emission-intensive grid electricity through the generation of low emission renewable electricity.	Displaces approximately 8,220 GWh of emission-intensive electricity through the generation of low emission renewable electricity over the project lifetime.
2)	13.0	Tons of greenhouse gas emissions avoided or removed	The total GHG emissions of the country will be reduced by the ERs and REMs that will no longer be contributing to the country's total GHG emissions. It is expected that the number of GHG emissions will be reduced annually as the project will be verified, confirming that these are no longer emitted to the atmosphere.	Displaces approximately 830,000 tCO _{2e} per annum of emission intensive grid electricity through the generation of low emission renewable electricity.	Displaces approximately 8,300,000 tCO _{2e} of emission-intensive grid electricity through the generation of low emission renewable electricity over the 10-year project lifetime.

3)	8.5	Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all	With total unemployment in South Africa having risen from 24.9% to 32.6% over the past decade (2014 to 2024) [8], this SDG target remains a high priority for the country. The country has a target to, “By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.”	The operational phase will provide 234 direct job opportunities.	The project planned to create approximately 234 job opportunities during the operational phase. Over the 10-year operational period, a percentage of the revenue will be dedicated to local communities through socio-economic initiatives.
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1.19 Additional Information Relevant to the Project

Leakage Management

As per the applied methodology ACM0002 (Ver. 22.0), no leakage emissions are considered in the project activity.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

No further information submitted.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification

The approach adopted in the stakeholder identification process was to contact as many stakeholders as possible through several channels, which included:

1. Placing advertisements in a local newspaper, Gemsbok, in April 2025.
2. Placing a noticeboard at the site and at strategic public locations in the vicinity of the project with a request for feedback. This included local supermarkets, municipality buildings and site access.
3. Providing written and email notice to potential stakeholders, including surrounding landowners, organs of state and relevant authorities; and
4. Requesting potential stakeholders to recommend other potential stakeholders to include them as members of the stakeholder database (chain referral process).

The stakeholder register is continuously updated as new I&APs are identified and/ or requested to be included or excluded in the database as part of the engagement process. The stakeholder register is maintained centrally.

All stakeholder details were compiled into a database which includes contact details, dates and details of consultations, and a record of all issues raised, and

	responses provided. This database will be updated on an on-going basis. Authorities regarded as key stakeholders: - Northern Cape Department of Environment and Nature Conservation (DEANC) - Department of Agriculture, Forestry and Fisheries: (DAFF): Directorate Land Use and Soil Management - Northern Cape: Department of Agriculture - Department of Water Affairs (DWA): Deputy Director Lower Orange Water Management Agency (WMA) - South African Heritage Resources Agency - Northern Cape Provincial Heritage: Boswa ya Kapa Bokone - ZF Mgcawu District Municipality - !Kheis Local Municipality - Kai !Garib Local Municipality and - Eskom
Legal or customary tenure/access rights	The legal tenure of the land is private ownership, and a binding lease agreement is in place with the landowners, in order to establish the Scatec RMI Hybrid Cluster plant facilities on the land. A binding lease agreement between the Project and the landowner was entered on the 6th of June 2022.
Stakeholder diversity and changes over time	The project site lies within the !Kheis Local Municipality, and key stakeholders that were identified include government agencies, neighbouring landowners and residents, local communities, businesses and environmental NGOs. The stakeholders cover a diverse spectrum of all aspects, including race, gender and economic profile in the given area. The economic sector in the local municipality is mainly dominated by agriculture which provides 51.8% of jobs [9]

	however, poverty is prevalent in the area, with a recorded unemployment rate of 27.4%. The mining sector is the second-most dominant industry in the broader district municipality of ZF Mgcau that provides work for local people. The two dominant industries provide male-oriented employment opportunities, which causes an uneven distribution of jobs between genders. The Scatec RMI Hybrid Cluster Project has rolled out a training programme, providing skills that enable employment in the project for unskilled local community members.
Expected changes in well-being	There are no significant negative impacts anticipated on the well-being of stakeholders. The project's implementation has brought multiple positive impacts, such as improved local economic activity and job creation. Over the 10-year operational period, the local community will also benefit from a community trust with an equity share in the project life to benefit historically advantaged communities.
Location of stakeholders	The project site is located in the Groblershoop local community (!Kheis Local Municipality) , and stakeholders that are most likely to be impacted by the project reside in the surrounding towns and communities of: - Kenhardt - !Kheis Local Municipality - Kai !Garib Local Municipality
Location of resources	The Scatec RMI Hybrid Cluster Project facility is located on the remaining extent of Onder Rugzeer Farm 168, with connection points to the substation on the remaining extent of Portion 3 of Gemsbok Bult Farm 120 within the !Kheis Local Municipality, Northern Cape Province.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	3 August 2015 – April 2025 (and ongoing)
Stakeholder engagement process	An Environmental Impact Assessment [9] was conducted for the project activities in 2015 which consisted of the Public Participation Process (PPP) where interested and affected parties were identified as stakeholders in the project. A database was collated through the PPP and formed part of the database for the VCS Stakeholder Engagement. Stakeholders were contacted primarily via email to inform them about the carbon project, prior to the quarterly stakeholder meeting on the 17th of March 2025. Meetings were held at various locations (Kenhardt and Groblershoop). This provided stakeholders with an opportunity to raise any queries or concerns about the carbon project. The meeting was held in a culturally appropriate and inclusive manner, and communication was facilitated in the local languages: English, Afrikaans and Setswana. In addition to the emails and calls, a newspaper article will be published in the local newspaper outlet – Gemsbok – in April 2025, where stakeholders are encouraged to provide consent and comments to a dedicated email provided by the carbon project proponent. Notices were also placed in public areas inviting comment.
Consultation outcome	The stakeholders were adequately informed about the carbon project and the process to be followed for validation and verification of the project. The information was well received, and stakeholders showed support of the carbon project and its positive impacts for the communities which will be benefitting from it. There were no concerns received via the dedicated comments section, nor any issues raised during the in-person engagement at the quarterly meeting.

Ongoing communication	Ongoing communication will be via direct electronic communication to stakeholders recorded in the registered stakeholders' database, as well as notices placed in public locations in the surrounding communities. In addition, the comments section on the Anthesis website will remain active throughout the lifetime of the project. Furthermore, ongoing liaison and communication between the project owner and neighbouring communities is facilitated by a locally employed facilitator (CLO). Regular face-to-face meetings and monitoring visits will also be conducted by the project owner.
Stakeholder input	Inputs and concerns received from stakeholders during the EIA process were carefully considered and addressed in the EIA and the project implementation. No issues or comments received were directly related to the carbon finance or VCS validation and verification process, therefore no updates to the project design have been considered necessary. Stakeholder comments will continue to be monitored and considered for future revision to the project design, operation and implementation.

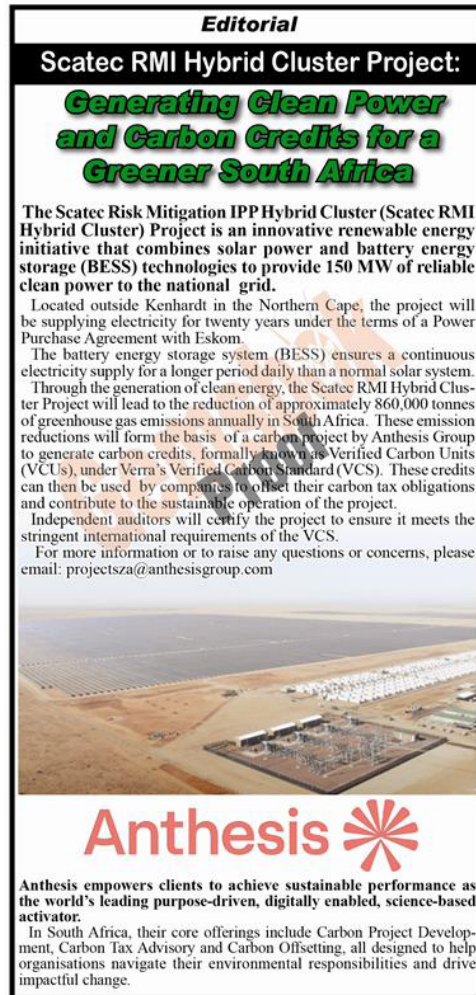


Figure 4: Article in "Gemsbok" - April 2025

2.1.3 Free Prior and Informed Consent

Obtaining consent

There is a legally binding land lease between the project site's landowners and the operating project proponent. No IPs (indigenous peoples), LCs (local communities) or customary rights holders are impacted by the Scatec RMI Project activities.

Outcome of FPIC	The outcome of the process is that the Scatec RMI Cluster Hybrid facility is established on private land for a period of 20 years with the option to renew. No other land resources or communities have been negatively impacted.
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2.1.4 Grievance Redress Procedure

Development process	The project proponent has developed a grievance mechanism to keep track of all grievances received from stakeholders. This consists of grievance forms and a grievance box located at the project site. In addition to this, stakeholders are given the platform to raise any concerns during the quarterly meetings held in their respective communities. A grievance register is kept to record all concerns and issues raised by the stakeholders via the communication channels that the project proponent has provided. Scatec also maintains a global grievance and whistleblower system that allows for anonymous raising of grievances that will be logged and handled on all issues raised (both internally and externally). In order to address grievances in a timely manner, the project proponent has set up a Grievance Committee that processes all stakeholder grievances received from the Community Liaison Officers stationed in the affected communities, as well as those submitted via the grievance box or recorded during the quarterly meetings. Anthesis has also provided a platform for any carbon project related grievances to be submitted via email or through a feedback form on the Anthesis or Scatec website.
Grievance redress procedure	Stakeholders have an ongoing opportunity to log into a dedicated feedback form on the Anthesis website in order to leave comments and engage with the project team. This website will be part of the ongoing and continuous engagement between the project and stakeholders. Comments received will be logged and taken into account in project operation and future verifications.

	Any grievances, complaints or conflicts from stakeholders would be captured via a dedicated feedback form on the Anthesis website, or from the other channels mentioned above. The Grievance Redress [10] process consists of: • Receiving the issue from a stakeholder • Assigning it to the relevant manager/team • Responding to the stakeholder (which may include telephonic, written or face to face communication) in an attempt to understand and resolve the issue in good faith • Recording the outcome of the interaction. • Grievances that cannot be resolved amicably in this manner will be referred to mediation by a neutral third party and failing resolution shall be raised to either arbitration or courts of law in South Africa.
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Grievances received	Resolution and outcome
Lack of employment opportunities for women during the construction phase.	A meeting was held to address the grievance, and the project proponent representative explained that although the construction phase had labour intensive job opportunities, more than 20% of the workforce on the project were women. The resolution was to identify more opportunities where female workers would be recruited during the operational phase.
Unfair distribution of job opportunities across the municipalities impacted by the project	The project proponents explained the recruitment process to the stakeholders and members of the !Kheis Local Municipality, detailing the closer proximity of the other municipalities which led to the uneven distribution of job opportunities.

2.1.5 Public Comments

Comments received	Actions taken
This section will be completed once the VCS public commenting period has passed. No comments received at the time of publishing.	This section will be completed once the VCS global public commenting period has passed.

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The Anthesis SA team has significant experience in implementing similar project activities, engaging with communities, project management and environmental and social safeguards aligned with the VCS requirements. Members of the project development team have as a collective, in excess of 50 years' experience in the voluntary carbon market. They have been originating, developing and monetizing carbon credits a company for over 2 decades. The Scatec RMI operational project proponent has extensive renewable energy project experience in the South African region and globally, including similar Renewable Power facilities in Brazil, Tunisia, Philippines, Honduras, Malaysia, Romania and Egypt among others. The integrated teams are well-positioned to implement and operate the facility and deliver the associated GHG emission reductions into the local market

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure taken
Natural and human-induced risks to stakeholders' wellbeing	No natural and human induced Risks identified	Measures to prevent natural and human-induced risks were taken and documented in the EIA report.
Risks to stakeholder participation	No risks identified	The stakeholder engagement process was conducted, and comments received from the stakeholders were properly addressed. No evidence of

		participation barriers to stakeholders identified.
Working conditions	No risks identified related to working conditions.	All workers and their working conditions are subject to the Basic Conditions of Employment Act of South Africa and labour conditions are covered by periodic inspections and compliance to the South African Department of Labour.
Safety of women and girls	No risks related to women and girls were identified.	The project proponent has taken all measures to ensure the safety of women and girls by implementing various policies as per South African labour and anti-discrimination laws.
Safety of minority and marginalized groups, including children	No risks were identified.	The project proponent ensures the safety of minority and marginalized groups, including children in the project as per the South African labour and anti-discrimination laws.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	Increased levels of air and noise pollution are anticipated during the construction phase due to higher traffic volumes.	Noise reduction actions must be implemented, such as use of baffle and noise screens on noisy machines. Dust suppression must take place on main access road; and continuous maintenance of gravel roads must be done to reduce air pollution from higher traffic volumes.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ⁴	Mitigation or preventative measure(s) taken
Discrimination	No risks identified	The project proponents have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa.
Sexual harassment	No risks identified	The project proponents have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa.
Equal pay for equal work	No risks identified	The project proponents have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa.
Gender equity in labor and work	No risks identified	The project proponents have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa.
Forced labor	No risks identified	The project proponents have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa.
Child labor	No risks identified	The project proponents have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa.

⁴ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

Human trafficking	No risks identified	The project proponents have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa.
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2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risks identified	The project proponent adheres to international and national human right laws and the Bill of Rights entrenched within the Constitution of South Africa. Risks related to human rights were assessed during the EIA process.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
Rock slab identified within the project site which is likely to be a grave since no other rocks are present in that vicinity.	The EMPr [3] incorporated mitigation measures that must be put in place prior to construction to exclude sites of significant cultural or historical (heritage) importance with a 5m buffer from the project boundary.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risks identified	All property rights are covered by legally binding lease agreements that cover land tenure, and no at-risk communities were involved or affected.

2.3.5 Benefit Sharing

As stated in Section 2.3.4, the project does not impact on property rights as the land is privately owned and has been secured via a lease agreement; therefore, there is no Benefit Sharing Agreement.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	- Disturbance of terrestrial fauna and flora on site due to construction workers and activities - Ousting of various fauna on account of long-term changes in the surrounding environment. - Mortality of medium and large terrestrial avifauna due to entrapment between perimeter fences.	The EMPr [11] recommends implementation of the following measures to mitigate the risks: - Conduct an Environmental Awareness Training and induction for all construction staff and personnel. - Identify points of excessive faunal activity and impact on operations. Undertake monitoring of faunal activities within the fenced area of the site and the immediate proximity of the site. - Facility to be fenced with a single perimeter fence or with two fences that are at least 4 metres apart.
Soil degradation and soil erosion	- Topsoil loss due to clearance of land surface during construction phase. - Soil erosion by wind or water due to alteration of the land surface characteristics.	The EMPr [11] addressed the risk of soil degradation by implementing the removal of at least 300mm of topsoil prior to construction, which was stockpiled and replaced into disturbed areas on completion of construction. In addition, erosion and sedimentation into water bodies must be minimised through effective stabilisation (gabions and

		Reno mattresses or similar) and re-vegetation of any disturbed riverbanks.
Water consumption and stress	Potential impact on groundwater availability due to use for cleaning solar panels, and groundwater availability as a result of accidental oil spillages and fuel leakages.	To reduce groundwater consumption, all buildings are to be fitted with rainwater collection and storage systems to supply water to taps and toilets, as well as any outdoor requirements such as cleaning of solar panels. Storage of hazardous/toxic liquids must take place within the contractors' site camp in an area approved by the ECO with a minimum distance of 60 metres away from any freshwater resources.

2.4.1 Rare, Threatened, and Endangered species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☒ Yes

☐ No

Species and habitat	Plant species that have been identified in the project site are the <i>Aloe dichotoma</i> and the <i>Aloe claviflora</i> which are protected under the provincial Northern Cape Nature Conservation Act. As part of the requirements for an Environmental Authorisation [12] to develop the project in this site, a permit was acquired to continue development with the implementation of a mitigation measure to map the tree species and exclude them from the project boundary.
Areas needed for habitat connectivity	The project site is located in an area that falls within the Bushmanland Arid Grassland which is poorly conserved, despite the presence of drainage lines and wetlands that

	provide habitats for riparian vegetation and fauna. The EMPr incorporated a sensitivity map to identify all the areas needed for habitat connectivity, which are to be excluded from the project development.
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	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	Loss of protected plant species <i>Aloe dichotoma</i> and <i>Aloe claviflora</i> that are present in the project site.	The EMPr [11] ensures that the project footprint does not include the protected plant species, and the appointed ECO will monitor the presence of the tree species in the project site.
Areas for habitat connectivity	Disruption of ecological processes within the drainage systems and wetlands, as well as loss of habitat for riparian vegetation and fauna.	Project development will exclude all drainage systems and wetlands present in the project site by use of a 32m buffer zone surrounding the hydrological features.

2.4.2 Introduction of species

The EMPr contains management interventions and mitigation measures for controlling alien invasive species that may occur on the site. No specie have been introduced as a direct part of the project activities.

The following existing alien species are located in the project area.

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
<i>Atriplex lindleyi</i>	- Pre-construction clearance of all exotic vegetation on site should be undertaken to reduce the possibility of further exotic weed invasion. - Design and implement an exotic weed control plan for the site for the duration of the operational phase.
<i>Datura ferox</i>	- Pre-construction clearance of all exotic vegetation on site should be undertaken to reduce the possibility of further exotic weed invasion. - Design and implement an exotic weed control plan for the site for the duration of the operational phase.

<i>Prosopis glandulosa</i>	- Pre-construction clearance of all exotic vegetation on site should be undertaken to reduce the possibility of further exotic weed invasion. - Design and implement an exotic weed control plan for the site for the duration of the operational phase.
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The table below identifies and summarizes any risks related to invasive species.

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species	A risk of establishment and spread of alien invasive plants due to the clearing and disturbance of indigenous vegetation exists	An exotic weed control plan was developed as part of the EMPr [12] to prevent introduction of invasive species. Additionally, vehicles used in the development site should be regularly washed to avoid the dispersal of seeds of any exotic or invasive species into the project site.

2.4.3 Ecosystem conversion

The project activity is not in any of the categories; ARR, ALM, WRC or ACoGS.

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	N/A	This project is not an ARR, ALM, WRC or ACoGS project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Table below includes the methodology(s), tools, and modules applied to the project activity.

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	ACM0002	Grid-connected electricity generation from renewable sources	22.0
Tool	TOOL 01	Tool for the demonstration and assessment of additionality	7.0
Tool	TOOL 07	Tool to calculate the emission factor for an electricity system	7.0
Tool	VT0011	Electricity System Emission Factors	1.0
Tool	TOOL 23	Methodological tool: Additionality of first-of-its-kind project activities	3.0

3.2 Applicability of Methodology

The project activity(s) meets each of the applicability conditions of the methodology(s), tools, and modules applied by the project (where applicable), as described below:

Methodology ID	Applicability condition	Justification of compliance
ACM0002	This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s);	a) The project activity involves construction and operation of grid-connected solar power plant integrated with BESS, at a site where no renewable energy power plant was operated prior to the implementation of the

Methodology ID	Applicability condition	Justification of compliance
	(c) Involve a retrofit of (an) existing operating plant(s)/unit(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s); or (f) Install a Greenfield power plant together with a grid-connected Greenfield pumped storage power plant. The greenfield power plant may be directly connected to the PSP or connected to the PSP through the grid.	project activity and therefore a “green field power project” as per the definition of the methodology and hence complies to the applicability condition (a). Hence the project activity meets the applicability condition of the methodology.
	In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic¹ or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s); (e) Integrate a BESS together with a Greenfield power plant that is operating in coordination with a PSP. The BESS is	This project activity is a grid-connected renewable energy power generation project which includes the integration of a BESS into the greenfield solar power plant. So, the project meets applicability condition (a)

Methodology ID	Applicability condition	Justification of compliance
	located at site of the greenfield renewable power plant.	
	The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit must have started commercial operation prior to the start of a minimum historical reference period of five years. The reference period is used for the calculation of baseline emissions and defined in the baseline emission section. Furthermore, no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity; (c) In case of Greenfield project activities applicable under paragraph 7(a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g., by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the	(a) This project is a hybrid renewable energy plant utilising solar power and wind power. (b) This project is not a capacity addition, retrofit, rehabilitation or replacement, but a greenfield plant. (c) BESS is an integral part of the project design and is a requirement of the project approvals and power purchase agreement. (d) BESS will only be charged from the greenfield

Methodology ID	Applicability condition	Justification of compliance
	associated renewable energy power plant(s). Only during exigencies² may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g., week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period. (e) In case the project activity involves PSP, the PSP shall utilize the electricity generated from the renewable energy power plant(s) that is operating in coordination with the PSP during pumping mode.	renewable energy (solar) plants at each location. No grid power will be used to charge the BESS component. (e) Not applicable to this project as the project does not employ PSP
	In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density,	Not applicable to this project activity as it does not employ hydro power.

Methodology ID	Applicability condition	Justification of compliance
	calculated using equation (7), is greater than 4 W/m²; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², and all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² are: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	
	In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or	Not applicable to this project as the project does not employ hydro power

Methodology ID	Applicability condition	Justification of compliance
	(b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	
	In the case of PSP, the project participants shall demonstrate in the VCS Project Description that the project is not using water which would have been used to generate electricity in the baseline.	Not applicable to this project as the project does not employ PSP
	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	(a) Not applicable to this project as the project does not employ any fuel switching. (b) Not applicable to this project as the project does not employ any biomass firing

Methodology ID	Applicability condition	Justification of compliance
	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	Not applicable to this project as the project does not employ any retrofits, rehabilitations, replacements, or capacity additions

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 01	The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	Since the applied methodology is not a new methodology, project proponent has applied this tool for the demonstration additionality in compliance with the tool. Refer to section 3.5 below for the detailed applicability of this tool and additionality assessment. Hence this tool is applicable.
CDM TOOL 01	Once the additionality tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	In line with the methodology requirement PP has applied this tool for the demonstration of additionality assessment. Hence this tool is applicable.

Methodology ID	Applicability condition	Justification of compliance
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CDM TOOL 07	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	This tool is applied to estimate the OM, BM and/or CM for the project activity
CDM TOOL 07	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	The emission factor for the project electricity system is calculated for grid power plants only.
CDM TOOL 07	In case of CDM projects the tool is not applicable if the project electricity	The project activity is not a CDM project, however the project activity is located in a

	system is located partially or totally in an Annex I country.	non-Annex I country and hence the tool is applicable.
CDM TOOL 07	Under this tool, the value applied to the CO2 emission factor of biofuels is zero.	N/A as the project activity does not involve biofuels.

Methodology ID	Applicability condition	Justification of compliance
TOOL VT 0011	This tool is applied to calculate the CO2 emission factor of an electricity system whose dispatch and generation are affected by a project activity. It is globally applicable.	The tool is applicable as the dispatch and generation is affected by a project activity.
TOOL VT 0011	All applicability conditions of the most recent version of TOOL07 must be met, except for Applicability Condition 5, which is removed under this revision	As per the table above, all applicability conditions of the most recent version of TOOL07 are met.
TOOL VT 0011	This tool must be used in conjunction with the most recent version of TOOL07. All the procedures and requirements of TOOL07 must be applied unless this tool indicates otherwise.	The tool has been used in conjunction with CDM TOOL07, as per the applicability conditions above and in section 5.3 the Grid Emission Factor determination.

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 23	This methodological tool is applicable to project activities that wish to use the "first-of-its kind" approach to demonstrate additionality and that use versions of baseline and monitoring methodologies, or the "Tool for the demonstration and	The project activity applied CDM methodology ACM0002 and Tool 01, which allow using the "first-of-its-kind" approach for demonstrating additionality. Hence this tool

	assessment of additionality" or the "Combined tool to identify the baseline scenario and demonstrate additionality", which allow using the "first-of-its-kind" approach for demonstrating additionality.	is applicable to this project activity.
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3.3 Project Boundary

.As per the methodology ACM002 version 22, the project boundary includes the power plants, and all power plants connected physically to the electricity system that the project power plant is connected to.

The table below provides an overview of the emissions sources included or excluded from the project boundary for determination of baseline and project emissions.

Source		Gas	Included?	Justification/Explanation
Baseline	CO ₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to project activity.	CO ₂	Yes	Main emission source (Grid electricity)
		CH ₄	No	Excluded for simplification. Minor source of emissions. Conservative
		N ₂ O	No	
		Other	No	No other GHG emissions relevant.
Project	Greenfield solar photovoltaic electricity generation project activity	CO ₂	No	No CO ₂ emissions from project activity. Minimal taken as negligible
		CH ₄	No	Excluded. Minimal taken as negligible
		N ₂ O	No	Excluded. Minimal taken as negligible
		Other	No	No other GHG emissions
	Charging of BESS using electricity from the renewable energy plants	CO ₂	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.
		CH ₄	No	Not accounted as BESS will only be charged from the renewable energy

Source	Gas	Included?	Justification/Explanation
			generation plants on site therefore taken as negligible.
	N ₂ O	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.
	Other	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.

The below map, clearly shows the physical locations of the project activity based on the description provided in Section 1.12 (Description of the Project Activity) above.

The diagram the equipment, systems and flows of mass and energy is also included in figure 2 below.



Figure 5: Physical location of project activity

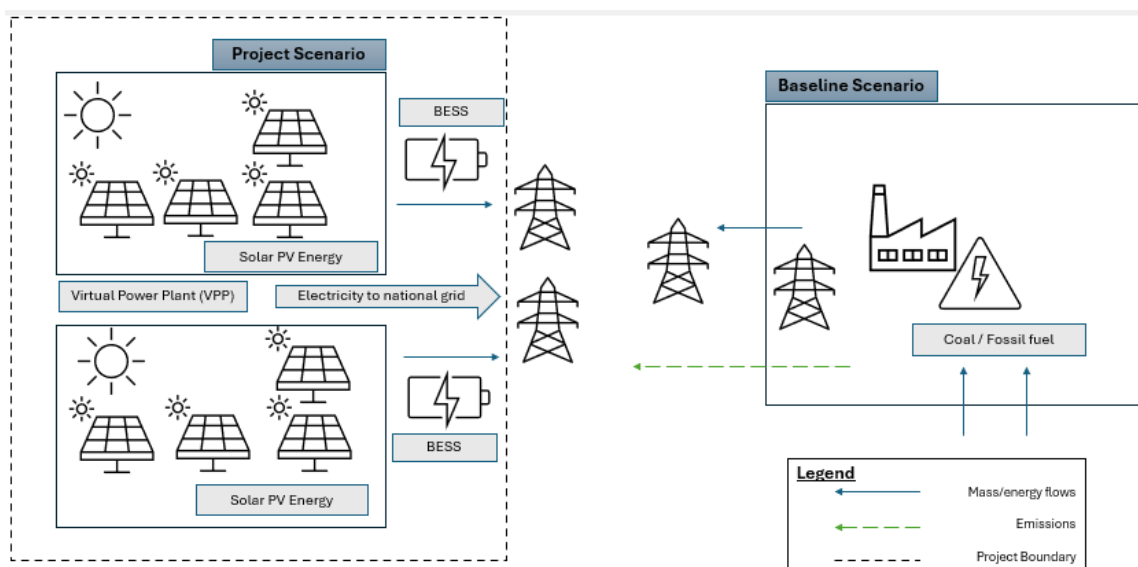


Figure 6: Project boundary, mass & energy flows of Scatec RMI Cluster Project

3.4 Baseline Scenario

The baseline scenario for the Scatec RMI Hybrid Cluster project is defined as the continuation of current grid operations, where electricity demand is primarily met by existing grid-connected fossil fuel (predominantly coal) power plants, and the addition of new fossil fuel based capacity/generation sources. This approach aligns with the requirements of the ACM0002 (Version 22.0) methodology, which states in Section 5.2.1: “If the project activity is the installation of a Greenfield power plant with or without Battery Energy Storage System (BESS), the baseline scenario is electricity delivered to the grid by the project activity that would otherwise have been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in AM TOOL07 ‘Tool to calculate the emission factor for an electricity system (Version 07.0)’.

Justification of the Baseline Scenario:

The South African electricity grid is characterized by high carbon intensity due to its heavy reliance on coal-fired power plants, which accounts for 80% of its energy mix [13]. This high reliance on coal positions South Africa among the top greenhouse gas emitters per capita globally [14].

According to South Africa’s Integrated Resource Plan (IRP 2019) [15] coal remains a dominant energy source, projected to account for over 50% of the electricity generation mix through 2030. According to South Africa’s Integrated Resource Plan (IRP 2019) [15], coal remains a dominant

energy source, projected to account for over 50% of the electricity generation mix through 2030. The IRP outlines significant reliance on existing coal-fired power plants, with limited expansion of renewable energy capacity due to financial, regulatory, and infrastructural barriers. Despite government efforts to increase renewable energy through initiatives like the Renewable Energy Independent Power Producer Procurement Programme (REIPPPP), the expansion pace is slow, and integration challenges persist, resulting in continued reliance on fossil fuel generation. The South African energy sector faces numerous challenges, including delays in project approvals, difficulties in securing financing, and grid connectivity issues. These barriers significantly impact the deployment of renewable energy projects, validating the assumption that the baseline scenario will continue to be dominated by fossil fuel-based power plants, including both existing and new coal and gas-fired units.

Historical data from Eskom, the primary electricity supplier, indicates a stable grid emission factor of approximately 0.96 tCO₂e/MWh over the past five years, driven by a heavy reliance on coal-fired plants. This trend is expected to continue in the absence of major interventions like the de Aar Hydra project, further supporting the selection of this baseline scenario. The grid emission factor for this baseline is calculated in accordance with the CDM TOOL07 “Tool to calculate the emission factor for an electricity system” Version 07.0. Option A under the Simple OM approach was selected along with the default weighting for solar projects in the first crediting period. This results in a combined margin factor for the baseline grid of 0.96 tCO₂e/MWh. The steps taken to calculate the grid emission factor as per AM TOOL 07 are outlined in section 4.1 below. §28 – 34 of ACM0002 do not apply to this project, as shown in Table 4 below:

Table 5: Identification of Baseline scenario as per ACM0002

Section number in ACM0002	Applicable to	Applicability to the project
5.2.2	Capacity addition to an existing renewable energy power plant or integration of a BESS to an existing solar photovoltaic or wind power plant/unit	N/A – the project is a Greenfield facility and not a capacity addition to an existing renewable energy power plant. The BESS components are also greenfield and exclusive to the greenfield De Aar Hydra generation and storage facilities.

5.2.3	Baseline scenario for retrofit or rehabilitation or replacement of an existing power plant	N/A – the project is a Greenfield facility and not a retrofit, rehabilitation or replacement of an existing power plant.
5.2.4	Baseline scenario for retrofit of an existing solar photovoltaic or wind power plant/unit with the integration of BESS	N/A – the project is a Greenfield facility and not a retrofit of an existing facility.
5.2.5	Baseline scenario for pumped storage projects operating in coordination with the renewable energy plants	N/A - the project is a Greenfield facility and is not connected to a Greenfield pumped storage power plant.

The project activity is the installation of a greenfield solar PV power plant with associated BESS component. The baseline scenario is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in TOOL07.

3.5 Additionality

3.5.1 Regulatory Surplus

Is the project registered or seeking registration in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country

☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes

☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes

☐ No

N/A – the project is located in the Republic of South Africa which is a Non-Annex 1 country and project activities are not mandated by a law, statute, or other regulatory framework.

3.5.2 Additionality Methods

Additionality is demonstrated by proving that the Project is first-of-its kind in the geographic area of South Africa. The following tools were used to evaluate additionality:

- CDM TOOL 01 version 7.0 Methodological Tool for the demonstration and assessment of additionality
- CDM TOOL 23 version 3.0 Methodological Tool for assessing additionality of first-of-its-kind project activities

CDM TOOL 01 version 7 prescribes a stepwise approach for evaluating additionality, and if Step 0 is satisfied the project is additional:

Step 0: Demonstrate whether the proposed activity is the first-of-its-kind.

This project activity is considered a first-of-its-kind project, as it is the first large scale project using BESS technology , powered by Solar PV, to be implemented within the country.

CDM TOOL 23 version 3.0 describes the first-of-its-kind approach for demonstrating additionality, and justification for these requirements is shown in the table below:

TOOL 23. Identification of a first-of-its-kind project activity	Justification of compliance
1. The project is the first in the applicable geographical area that applies a technology that is different from technologies that are implemented by any other project, which are able to deliver the same output and have started commercial operation in the applicable geographical area before the project design document [<i>Project Description (PD)</i>] is published for global stakeholder consultation or before the start date of the proposed project activity, whichever is earlier.	i. The applicable geographic area is the entire host country (i.e. Republic of South Africa); ii. The proposed project activity is the first project of this scale (150MW collective contracted capacity) using BESS technology, powered by solar PV technology to reach commercial operation in South Africa Other projects that apply the same combination of technologies are shown in Table 4 and their outputs range from 75MW to 150MW contracted capacity. While the smallest was a factor of 50% lower and none exceeded the project capacity. They were all considered in the comparison iii. Only the Scatec RMI Hybrid Cluster had reached commercial operation at the time

	of this Validation, as it reached initial COD on 18 November 2023 [1]and achieved completed COD for the entire project capacity on 8 December 2023. This is earlier than any of the other comparable projects. Evidence to support the above claims is publicly available, as shown below.
2. Measure - project implements one or more of the following four emission reduction measures: (a) Fuel and feedstock switch (example: switch from naphtha to natural gas for energy generation, or switch from limestone to gypsum in cement clinker production); (b) Switch of technology with or without change of energy source including energy efficiency improvement as well as use of renewable energies (example: energy efficiency improvements, power generation based on renewable energy); (c) Methane destruction (example: landfill gas flaring); (d) Methane formation avoidance (example: use of biomass that would have been left to decay in a solid waste disposal site resulting in the formation and emission of methane, for energy generation).	i. The proposed project activity uses renewable energy to charge the BESS, therefore is allowable under Tool 23 2.(b) and battery storage/BESS is not excluded under the VCS Standard version 4.7. ii. This project does not include industrial gases, transport and afforestation/reforestation projects – which are not allowed under Tool 23.
3. Output - goods/services produced by the project activity includes, among other things, heat, steam, electricity, methane, and biogas unless otherwise specified in the applied methodology.	i. Electricity generated from renewable sources like solar PV, is allowable under Tool 23, and battery energy storage systems are not excluded under the VCS Standard version 4.7.
4. Different technologies - technologies deliver the same output and differ by at least one of the following: (a) Energy source/fuel (e.g. energy generation by different energy sources such as wind and hydro and different	i. The proposed project activity is a large scale project, as the BESS is to have a combined energy capacity of 495 MWh, with 225 MW dispatch capacity at the Delivery Point and 150 MW contracted capacity; ii. The BESS is charged from the solar PV plant, split over three footprints located adjacent to one another and forming a

types of fuels such as biomass and natural gas); (b) Feed stock (e.g. production of fuel ethanol from different feed stocks such as sugar cane and starch, production of cement with varying percentage of alternative fuels or less carbon-intensive fuels); (c) Size of installation (power capacity) or energy savings: (i) Micro (<5MW); (ii) Small (<15MW); (iii) Large (>15MW)	combined total installed capacity of an approximately 540 MW peak. iii. A comparison with other projects awarded ‘preferred bidder’ status under the Risk Mitigation Independent Power Producer Procurement Programme (RMIPPPP) Bid Submission process, conducted from August 2020 until 1 June 2021 is shown in Table 4 (below). From this List of Additional Preferred Bidders, Scatec Kenhart cluster was the first to reach commercial operation, and uses similar BESS with RE technologies as the other projects (with the exception of Mulilo Total Coega (a Solar PV project only) and the three Karpowership projects (proposed as gas fired floating power plant). The RMIPPPP process is the only process in the host country that allows for projects of this scale to supply energy to the national grid and no previous or future rounds have been repeated at the time of this Validation.
5.First-of-its-kind project activities select a crediting period that is “a maximum of 10 years with no option of renewal”.	i. The start and end dates of the 10-year fixed crediting period are 18-November-2023 to 17-November-2033.

Evidence to demonstrate this project activity is the first-of-its-kind for BESS powered by renewable energy, is found on the REIPPP webpage. The South African government, via it’s Department of Mineral Resources and Energy (DMRE), National Treasury (NT) and the Development Bank of Southern Africa (DBSA) established the IPP Office for the specific purpose of transparent reporting and delivering on the Independent Power Producers Procurement Programme (IPPPP) objectives⁵. From the Risk Mitigation Project section of the Independent Power Producers (IPP) Office website⁶, the following projects are listed:

⁵ <https://www.ipp-projects.co.za/Home/About>

⁶ <https://www.ipp-rm.co.za/>

Project name	Technology	Contract capacity (MW)	Commercial Operation Status	Comments
Scatec RMI Hybrid Cluster	BESS (Solar PV)	150	18 November 2023	First project using this technology and of this scale to start commercial operation (VCS project # 5488)
Oya Hybrid	BESS (Solar PV & Wind)	128	Planned operation in future (October 2025)	Still under construction (VCS project # 5312)
ACWA Power DAO	BESS (Solar PV)	150	Planned operation in future (Q2 2026 ⁷)	Project signed PPA in December 2023 with planned operation scheduled for Q2 2026. Construction had not commenced at the time of this assessment.
Umoyilanga Energy	BESS (Solar PV & Wind)	75	Planned operation in future (October 2025)	Still under construction (VCS project # 5272)
Karpowership SA Coega	Gas fired floating power plant	450	On hold ⁸	Electricity and Energy Minister Kgosietsso Ramokgopa announced deal was no longer viable in October 2024. Technology does differ from project activity.
Karpowership SA Richards Bay	Gas fired floating power plant	450	On hold ⁹	Electricity and Energy Minister Kgosietsso Ramokgopa announced deal was no longer viable in October 2024. Technology does differ from project activity.

Karpowership SA Saldanha	Gas fired floating power plant	450	On hold ¹⁰	Electricity and Energy Minister Kgosiemo Ramokgopa announced deal was no longer viable in October 2024. Technology does differ from project activity.
Mulilo Total Coega	Solar PV	198	Construction has not started yet ¹¹	Project has been renamed and sold and continue with new ownership ¹² . Technology is solar PV only and construction had not commenced at the time of assessment.
Mulilo Total Hydra Storage	BESS (Solar PV)	75	Planned operation in future (September 2025)	Project has been renamed and sold and continue with new ownership ¹³ . Still under construction (VCS project # 5337).

Table 6: List of preferred bidders announced in March and June 2021 by IPP Office

Further to this, as of August 2024, the National Energy Regulator of South Africa (NERSA), lists all section 34 IPPs that have been awarded contracts under the REIPPP programme. [16] Only 7 BESS projects are listed which includes the 3 portions of the Scatec RMI Hybrid Cluster project. [16] Only 7 BESS projects are listed which includes the 3 portions of the Scatec RMI Hybrid Cluster project. These

⁷ <https://www.acwapower.com/news/acwa-power-signs-power-purchase-agreement-for-south-africas-largest-hybrid-dispatchable-renewable-energy-project/>

⁸ <https://www.engineeringnews.co.za/topic/energy>

⁹ <https://www.engineeringnews.co.za/topic/energy>

¹⁰ <https://www.engineeringnews.co.za/topic/energy>

¹¹ <https://www.power-technology.com/data-insights/power-plant-profile-mulilo-total-coega-solar-pv-park-south-africa/>

¹² <https://www.mulilo.com/news/2023/mulilo-energy-holdings-risk-mitigation-ipp-procurement-programme/>

¹³ <https://www.mulilo.com/news/2023/mulilo-energy-holdings-risk-mitigation-ipp-procurement-programme/>

projects are all already included in Table 4 above and this therefore provides further corroborating evidence to show the Scatec RMI Hybrid Cluster was the first to reach commercial operation, and is therefore the first-of-its-kind in South Africa.

110	REIPPP BWS	RE_PV_0037	Grootfontein PV 3	Grootfontein PV 3 (RF) Proprietary Limited	SOLAR PV
111	RMIPPP	RM-TA-0025	ACWA Power Project DAO	ACWA Power Project DAO (RF) Proprietary Limited	Solar PV (BESS)
112	RMIPPP	RM-TA-0028	Umoyilanga Energy	Umoyilanga (RF) Proprietary Limited	Hybrid (BESS)
113	RMIPPP	RM-TA-0180	TotalEnergies Hydra Storage	Total Energies Hydra Storage (RF) Proprietary Limited	Solar PV (BESS)
114	RMIPPP	RM-TA-0032	Scatec Kenhardt 1	Scatec Kenhardt 1 (RF) Proprietary Limited	Solar PV (BESS)
115	RMIPPP	RM-TA-0195	Scatec Kenhardt 2	Scatec Kenhardt 2 (RF) Proprietary Limited	Solar PV (BESS)
116	RMIPPP	RM-TA-0195	Scatec Kenhardt 3	Scatec Kenhardt 3 (RF) Proprietary Limited	Solar PV (BESS)
117	RMIPPP	RM-TA-0006	Oya Energy Hybrid Facility	Oya Energy Proprietary Limited	Solar PV (BESS)
118	DOE PEAKERS	N/A	Avon	Avon Peaking Power (RF) (Pty) Ltd	Open Cycle Gas Turbine

Figure 7: Extract from NERSA Annex 1 list of IPP projects - BESS Technology only

Further to this, a project search was conducted (including during the development of a national Grid Emission Factor using TOOL 07 which considers CDM, VCS and Gold Standard plants in the Build Margin Calculation (Table 12). This confirmed that none of these plants applied a similar technology for dispatchable power so were not considered in the determination of this step of additionality.

Further to this in the presence of Minister of Mineral Resources and Energy, Gwede Mantashe; Acting Director General of the Department of Mineral Resources and Energy, Patricia Gamede; Eskom CEO Andre de Ruyter, Head of the IPP Office, Bernard Tshifhiwa Magoro stated that the, “Scatec projects signed on 02 June 2022, are a **“first of its kind”** in South Africa, combining Solar PV and Battery technologies to contribute 150MW of dispatchable capacity to the national grid.” [17] Supplemental to this the Department of Mineral Resources and Energy announced in July 2022 that, “Once completed, the (..) power stations will be the first dispatchable renewable energy projects in South Africa.” [2] [More recently it was confirmed that the hybrid project was, “one of the world’s **first** and largest hybrid solar and battery storage facilities.” [18]

The conclusion of this step is that the project has demonstrated that it is a first-of-its-kind activity and as per §13 of TOOL 23 and §4.1.18 of TOOL 01 the project is considered additional and it is not required to proceed to steps 2, 3 or 4.

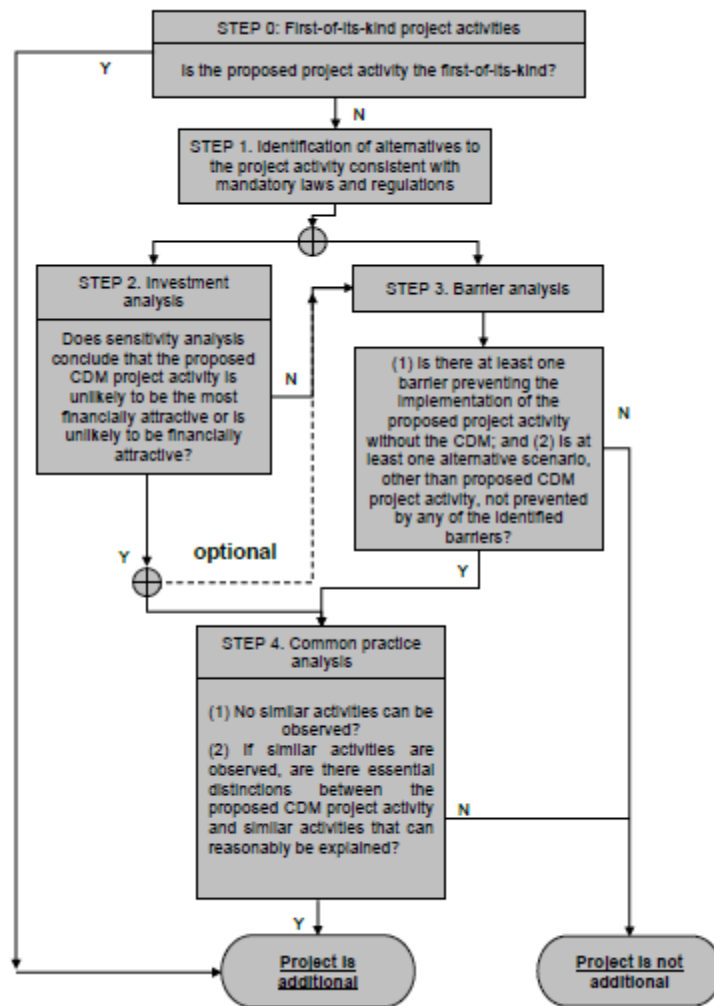


Figure 8: Flowchart of the step-wise additionality approach from TOOL 01

3.6 Methodology Deviations

There are no methodology deviations applied in the Project.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

The Project achieved its Commercial Operation Date (COD) with the signature of the Notice of Commencement of the Facility between the facility operating entity and the national power utility formally commencing the Power Purchase Agreement. This was achieved for the first portion of the project on 18 November 2023 [6]. This was the start date of the project and GHG reduction activity.

From this date, the plant operated and the second portion of the project achieved COD on 25 November 2023 [19] and the remaining portion on 8 December 2023 [20]. No other changes from the project implementation outlined in this document were experienced in this initial verification period.

While the plant experienced normal operations during the monitoring period, the following abnormal incidents were recorded and managed during the period.

Incident	Impact
Grid event – Eskom overhead line & current transformer (CT) failure.	Reduction of a portion of plant supply while incident was repaired
Setpoint instability after initial grid events.	OEM intervention and faults rectified with preventive actions. Limited production loss.
Overhead line pole failure.	Limited production loss while incident was repaired.
Power Plant Controller (PPC) set point instability.	OEM intervention and faults rectified with preventive actions. Minimal production loss.
Lightning strikes on overhead lines.	Continued risk in thunder storms. Each incident resulted in limited production loss while incident was repaired

BESS software issues.	OEM intervention and faults rectified with preventive actions and local capacity knowledge transfer. Limited production loss.
Power conversion system (PCS) overheating.	Root cause analysis intervention and faults rectified with preventive actions including planned maintenance. Limited production loss.

5 Quantification of Estimated GHG Emission Reductions and Removals

5.1 Baseline Emissions

The baseline emissions were quantified in accordance with the methodology ACM0002 version 22.0. For this project, it includes electricity delivered to the South African National grid. This was calculated as per the calculation steps described below.

Baseline Emission Calculation

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Equation 1 (ACM0002 Equation 11)

Where:

BE_y	=	Baseline emissions in year y (tCO ₂ /yr)
$EG_{PJ,y}$	=	Net electricity fed into the grid supplied by the project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	=	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using TOOL07 (t CO ₂ /MWh)

The results obtained are: 830 000 t CO₂/ annum, as shown below:

Parameter name	Value	Unit
BE_y	830, 000	t CO ₂ / yr
$EG_{PJ,y}$	860, 000	MWh/yr
$EF_{grid,CM,y}$	0.96	t CO ₂ /MWh

The plant's auxiliary electricity consumption is supplied by its connection to South Africa's national grid (Eskom).

$EG_{PJ,y}$ represents the net electricity supplied to the grid, which is measured as the difference between electricity exported to the grid and electricity imported from the grid. The plant does not rely on the on-site diesel generators for electricity generation, it only serves as emergency backup. Since the plant does not rely on the backup diesel generators to meet its auxiliary electricity supply needs, the emissions from such generator are considered negligible as per paragraph 42 of ACM0002. Fossil fuel usage will, however, still be monitored for record-keeping purposes.

The project is a greenfield power plant and therefore $EG_{PJ,y}$ is calculated as follows:

$$EG_{PJ,y} = EG_{facility,y}$$

Equation 2 (ACM0002 Equation 12)

Where:

$EG_{PJ,y}$ = Net electricity generation fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)

$EG_{facility,y}$ = Net electricity generation supplied by the project plant to the grid in year y (MWh/yr)

The results obtained are: 860, 000 MWh / yr (from technical specifications)

Grid Emission Factor Calculation

The grid emission factor for South Africa's national electricity grid was calculated in accordance with the following tools:

- CDM Tool 07 version 7.0 [21]CDM Tool 07 version 7.0 [21]CDM Tool 07 version 7.0 [21]CDM Tool 07 version 7.0 [21]
- VCS Tool VT0011 version 1.0 [22]VCS Tool VT0011 version 1.0 [22]VCS Tool VT0011 version 1.0 [22]VCS Tool VT0011 version 1.0 [22]

According to these tools, the combined margin (CM) grid emission factor is calculated as the weighted average of the system's operating margin (OM) and build margin (BM). The operating margin refers to the group of existing power plants whose current electricity generation would be affected by the proposed renewable energy project activity. The build margin represents the likely future power plants in the electricity systems whose construction and future operation would be affected by the proposed renewable energy project activity. This calculation is only applicable to grid-connected power plants and does not include off-grid power plants.

In accordance with Tool07 The following 6 steps are followed:

- **Step 1:** Identify the relevant electricity systems
- **Step 2:** Choose whether to include off-grid power plants in the project electricity system (optional)
- **Step 3:** Select a method to determine the operating margin (OM)
- **Step 4:** Calculate the operating margin emission factor according to the selected method
- **Step 5:** Calculate the build margin (BM) emission factor
- **Step 6:** Calculate the combined margin (CM) emission factor.

5.1.1 Step 1: Identify the project electricity system

The project electricity system was identified, using Option 2, as: The national grid of South Africa – administered by the national power utility, Eskom Holding SOC Ltd, including generation from Independent Power Producers (IPPs) and imports from the South African Power Pool (SAPP) network. This is shown schematically in Figure 12 below.

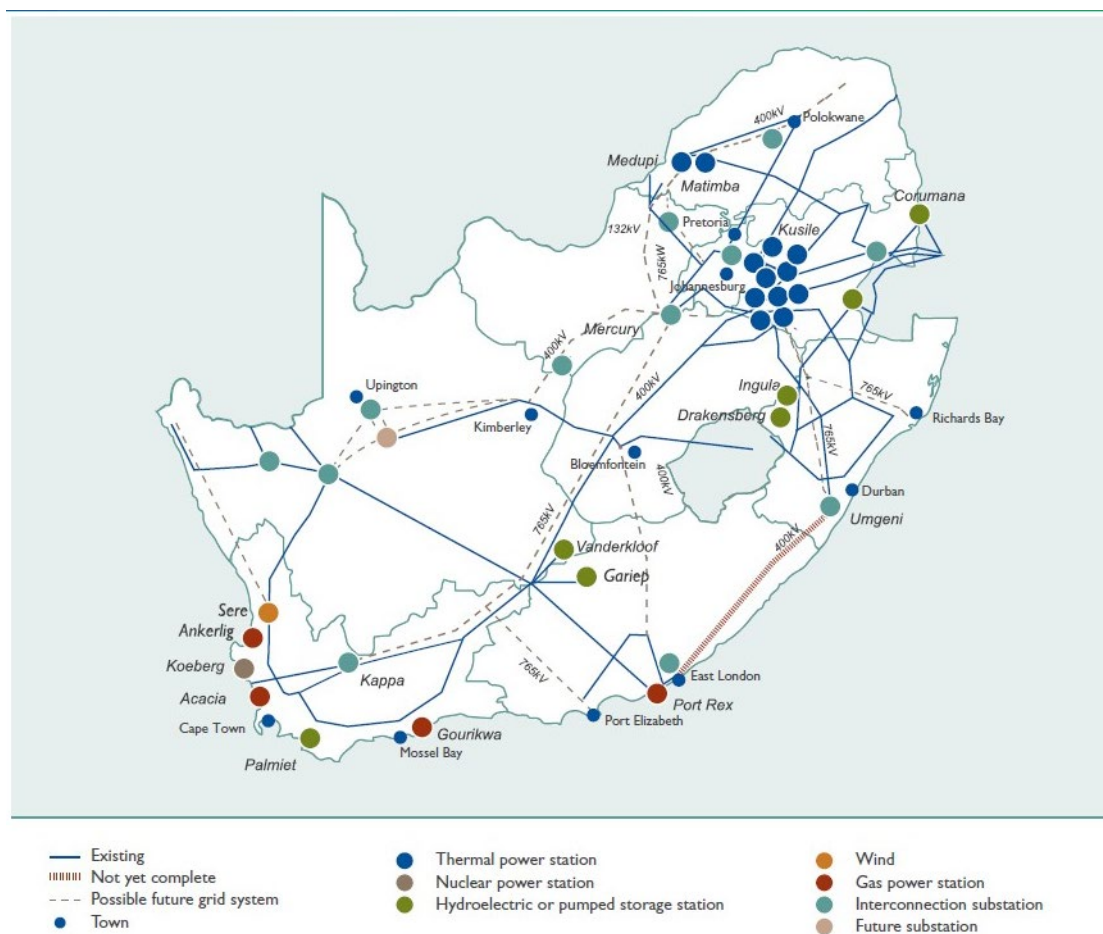


Figure 9: Project electricity system Source: [13]

South Africa's electricity grid is not connected to an electricity system located partially or totally in an Annex I country.

5.1.2 Step 2: Choose whether to include off-grid power plants in the project electricity system

Only grid connected plants (shown in Table 10). were included in this calculation.

5.1.3 Step 3: Operating Margin method selection

The operating margin (OM) emission factor ($EF_{grid,OM,y}$) was calculated based on the simple OM method (paragraph 38.a in TOOL07). This was decided based on the following criteria:

- The share of electricity supplied by Low-Cost/Must-Run (LCMR) plants in the last 5 years has ranged from 19% – 23%. It has been consistently lower than 50%, as calculated using TOOL07 40(a) Approach 1, by Equation 3 below:

$$Share_{LCMR} = average \left[\frac{EG_{LCMR(y-4)}}{total_{y-4}}, \dots, \frac{EG_{LCMR(y)}}{total_y} \right]$$

Equation 3 (Tool 07, 40 Equation 1)

Where:

$Share_{LCMR}$	=	Share of the low cost/must run resources (%)
EG_{LCMRy}	=	Electricity generation supplied by LCMR plants in year y (MWh)
$total_y$	=	Total electricity supplied to the project electricity system in year y (MWh)
Y	=	The most recent year for which data is available

The average share of LCMR plants ($share_{LCMR}$) was found to be equal to 23.6% over the last 5 years.

As per paragraph 45 of VT0011, power plants registered under the VCS program or any other GHG program are included in the sample group used to calculate the operating margin, not including wheeling agreements. As per paragraph 47(a) of tool07, option A was chosen based on the fact that net electricity generation and fuel usage data is available for each power plant serving the system.

The ex-ante option (TOOL07 paragraph 42(a)) was chosen to determine the simple OM. The 3-year generation-weighted average emission factor was calculated based on the most recent data available at the time of listing the project on Verra's registry. For this project, the most recent data available is 2024. Therefore, the generation weighted average of 2022, 2023, and 2024 was used.

The results obtained are shown in Table 6 below

Table 7: LCMR Analysis

Parameter name	Value	Unit
Share _{LCMR} : 2024	23.6%	Per cent
Share _{LCMR} : 2023	23.1%	Per cent
Share _{LCMR} : 2022	21.3%	Per cent
Share _{LCMR} : Average (5 years)	21.2 %	Per cent
EG _{LCMRy} : 2024	49 751	GWh / yr
EG _{LCMRy} : 2023	49 691	GWh / yr
EG _{LCMRy} : 2022	48 092	GWh / yr
total _y : 2024	210 648	GWh / yr
total _y : 2023	215 318	GWh / yr
total _y : 2022	226 226	GWh / yr

5.1.4 Step 4: Operating margin calculation

The simple OM emission factor was calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including LCMR power plants. The following plants were identified as LCMR, and therefore excluded from the OM calculation:

Table 8 LCMR plants serving the project electricity system

LCMR Plant name	Technology	Reason for classification as LCMR
- Gariep - Vanderkloof - Mbashe - First Falls - Ncora - Second Falls	Hydroelectric power stations	Renewable (low generation costs)
- Drakensberg - Ingula - Palmiet	Pumped storage stations	Only used during periods of peak demand (must run)
Sere	Wind energy stations	Renewable (low generation costs)
Koeberg	Nuclear power stations	Nuclear listed as LCMR
- Biomass - Concentrated solar power	Imports and REIPPP plants	Imports or renewable

• Gas/liquid fuel • Hydroelectric • Landfill • Solar PV • Wind		
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The power plants used in the OM calculation are shown in Table 10 below. The simple OM was calculated based on the net electricity generation and the CO₂ emission factor of each power unit (TOOL07 paragraph 47, option A), as per Equation 4.

$$EF_{grid,OMsimple,y} = \frac{\sum m EG_{m,y} \times EF_{EL,m,y}}{\sum m EG_{m,y}}$$

Equation 4 (Tool 07 48, Equation 3)

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO ₂ emission factor in year y (t CO ₂ /MWh)
$EG_{m,y}$	=	Net electricity delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO ₂ emission factor of power unit m in year y (t CO ₂ /MWh)
m	=	All power units serving the grid in year y except low-cost/must-run power units
y	=	The relevant year as per the data vintage chosen

The results obtained are: $EF_{grid,OMsimple,y} = 1.1158$ tCO₂/MWh

All values used are shown in Table 10.

Table 9: Operating Margin Plant Data

Plant Name	Electricity generated (MWh/yr) 2022	Electricity generated (MWh/yr) 2023	Electricity generated (MWh/yr) 2024	Emission factor 2023 (tCO ₂ /MWh)	Emission factor 2022 (tCO ₂ /MWh)	Emission factor 2021 (tCO ₂ /MWh)
Arnot	9 085 064	7 530 158	7 530 158	1.17	1.16	1.14
Duvha	9 822 489	9 555 266	9 555 266	1.11	1.13	1.13
Hendrina	4 738 179	2 136 620	2 136 620	1.35	1.24	1.20
Kendal	15 139 746	15 750 630	15 750 630	1.02	1.05	1.04
Kriel	12 597 519	10 727 877	10 727 877	1.32	1.28	1.22
Lethabo	22 890 481	21 367 191	21 367 191	1.00	0.97	0.95
Matimba	23 180 336	23 257 630	23 257 630	1.02	1.04	0.99
Majuba	19 823 301	15 540 628	15 540 628	1.19	1.13	1.09
Matla	18 412 664	15 170 449	15 170 449	1.30	1.27	0.99
Tutuka	8 524 715	7 564 337	7 564 337	1.18	1.23	1.22
Camden	6 670 341	7 378 745	7 378 745	1.10	1.16	1.25
Grootvlei	2 445 162	2 416 293	2 416 293	1.21	1.27	1.24
Komati	608 728	369 330	369 330	1.01	1.03	1.06
Kusile	7 992 189	9 119 476	9 119 476	1.01	1.30	1.09
Medupi	22 637 675	23 246 212	23 246 212	0.88	0.92	0.91
Acacia	873	1 069	1 069	1.17	1.16	1.14
Port Rex	937	3 010	3 010	1.17	1.16	1.14
Ankerlig	953 788	1 332 491	1 332 491	1.17	1.16	1.14
Gourikwa	870 396	1 681 217	1 681 217	1.17	1.16	1.14
Total	186 394 583	171 134 921	174 148 629			

The emission factor of each power unit ($EF_{EL,m,y}$) was determined following option A1 (paragraph 49.a) as shown in Equation 5 below:

$$EF_{EL,m,y} = \frac{\sum_i FC_{i,m,y} \times NCV_{i,y} \times EF_{CO_2,i,y}}{\sum EG_{m,y}}$$

Equation 5 (Tool 07 49.(a) Equation 4)

Where:

$EF_{EL,m,y}$	=	CO ₂ emission factor of power unit m in year y (t CO ₂ /MWh)
$FC_{i,m,y}$	=	Amount of fuel type i consumed by power unit m in year y (Mass or volume unit)
$NCV_{i,y}$	=	Net calorific value of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO_2,i,y}$	=	CO ₂ emission factor of fuel type i in year y (t CO ₂ /GJ)
$EG_{m,y}$	=	Net quantity of electricity delivered to the grid by power unit m in year y (MWh)
m	=	All power units serving the grid in year y except low-cost/must-run power units
i	=	All fuel types combusted in power unit m in year y
y	=	The relevant year as per the data vintage chosen

The emission factor for each power unit ($EF_{EL,m,y}$) is shown in Table 10 above.

The values used in Equation 5 are shown in Table 11 below.

$EG_{m,y}$ was obtained from Eskom's website as well as their annual Integrated Reports. [13], [23].

Table 10: Parameters for plant emission factor calculation

Plant Name	Coal usage (t) 2022	Coal usage (t) 2023	Coal usage (t) 2024	Fuel oil usage (t) 2022	Fuel oil usage (t) 2023	Fuel oil usage (t) 2024	Net calorific value coal (GJ/t)	Net calorific value fuel oil (GJ/t)	Emission factor coal (tCO ₂ /GJ)	Emission factor fuel oil (tCO ₂ /GJ)
Arnot	5 186 766	4 293 316	4 828 094	37 792.2	45 179.7	46 635	19.29	43	0.09	0.08
Duvha	5 732 195	5 428 210	6 802 842	39 550.9	71 179.4	72 479	19.29	43	0.09	0.08
Hendrina	3 024 981	1 486 626	1 426 720	11 012.5	7 722.1	9 262	19.29	43	0.09	0.08
Kendal	9 362 475	9 395 998	7 712 146	61 174.9	106 402.9	113 212	19.29	43	0.09	0.08
Kriel	7 838 445	6 865 127	5 183 313	39 516.2	53 949.1	56 540	19.29	43	0.09	0.08
Lethabo	16 277 939	15 646 686	16 048 047	12 792.0	14 075.3	13 252	19.29	43	0.09	0.08
Matimba	13 430 751	13 179 578	10 674 314	9 944.3	14 066.5	18 715	19.29	43	0.09	0.08
Majuba	11 924 693	9 831 151	11 304 146	149 002.7	140 169.2	19 808	19.29	43	0.09	0.08
Matla	10 842 908	9 536 046	10 147 328	14 520.0	15 923.0	19 578	19.29	43	0.09	0.08
Tutuka	5 510 214	4 697 013	3 879 070	68 779.3	64 592.8	62 137	19.29	43	0.09	0.08
Camden	3 932 224	4 129 344	4 402 710	57 014.4	49 016.8	56 731	19.29	43	0.09	0.08
Grootvlei	1 544 269	1 518 602	1 064 085	61 318.1	21 199.1	12 594	19.29	43	0.09	0.08
Komati	335 833	197 954	-	1 332.0	1 773.0	-	19.29	43	0.09	0.08
Kusile	4 783 403	5 035 930	5 481 821	534 043.2	10 351.0	26 640	19.29	43	0.09	0.08
Medupi	11 351 788	11 181 866	11 065 548	22 637.7	23 246.2	20 489	19.29	43	0.09	0.08

Plant Name	Diesel usage (l) 2022	Diesel usage (l) 2023	Diesel usage (l) 2024	Net calorific value Diesel (MJ/l)	Emission factor Diesel (tCO ₂ /GJ)
Acacia	350 693	414 371	936 177	35.5	0.074
Port Rex	364 144	1 093 406	1 458 236	35.5	0.074
Ankerlig	300 686 482	418 610 227	507 827 227	35.5	0.074
Gourikwa	269 555 564	515 526 621	622 893 869	35.5	0.074

5.1.5 Step 5: Calculate the build margin (BM) emission factor

An ex-ante approach (option 1 in TOOL07 paragraph 72(a)) is selected to calculate the BM emission factor. The most recent information available on units already built for sample group m at the time of submitting the project for validation was used. The procedure followed to determine the sample group of power units (m) used to calculate the build margin is described below. The total electricity generation (AEG_{total}) of all power plants directly supplying the system in 2024 was 184,576 GWh.

In line with VT0011, the set of five power units, including power units registered as VCS or other GHG programs, that started to supply electricity to the grid most recently was identified. Their annual generation was determined ($AEG_{SET-5 \text{ units}}$) as 1 643 833 MWh in 2024, which was below 20% of AEG_{total} .

Table 11: Five most Recent Plants supplying the grid

No.	Plant name	Commissioning Date	Electricity supplied UOM	2024
1	TWE Golden Valley Wind Power Project	2021/05/01	MWh	509 303
2	GS11949 Small-scale solar electrical programme, South Africa	2020/12/16	MWh	22 500
3	Solar Energy Programme for South Africa	2019/02/01	MWh	381 930
4	Longyuan Mulilo De Aar Maanhaarberg Wind Energy Facility	2017/10/31	MWh	290 500
5	Longyuan Mulilo De Aar 2 North Wind Energy Facility	2017/10/31	MWh	439 600
TOTAL				1 643 833

Next, the set of power units that started to supply power to the grid most recently and that comprise 20% of the total annual electricity generation of the electricity system (AEG_{total}) was identified ($SET_{>-20 \text{ percent}}$). This set is shown in Table 12 and included units from VCS or other GHG programs. The set's annual generation was determined ($AEG_{set>-20 \text{ percent}}$). This was 208 462 079 MWh in 2024).

$AEG_{set>-20 \text{ percent}}$ was found to be greater than $AEG_{set 5 \text{ units}}$, and therefore $SET_{>-20 \text{ percent}}$ was chosen as SET_{sample} .

The resulting SET_{sample} group of power units, m, used to calculate the build margin is shown Table 12.

Table 12: Plant set for BM Calculation

No.	Plant name	Commissioning Date	Power Produced in Year: 2024
Eskom Fleet Plants			
1	Gourikwa	2007/07/01	2 038 316 MWh
2	Medupi	2015/08/22	22 626 444 MWh
3	Kusile	2017/08/30	10 262 515 MWh
IPP Power Plants			
4	Green Power For South Africa	2013/05/31	1 219 826 MWh
5	South African Grid Connected Wind Farm Programme	2014/04/01	318 056 MWh
6	TWE Golden Valley Wind Power Project	2021/05/01	509 303 MWh
7	GS11949 Small-Scale Solar Electrical Programme, South Africa	2020/12/16	22 500 MWh
8	Solar Energy Programme For South Africa	2019/02/01	381 930 MWh
9	Longyuan Mulilo De Aar Maanhaarberg Wind Energy Facility	2017/10/31	290 500 MWh
10	Longyuan Mulilo De Aar 2 North Wind Energy Facility	2017/10/31	439 600 MWh
11	Amakhala Emoyeni Grid Connected 138.6 MW Wind Farm, Phase 1, South Africa	2016/06/15	450 559 MWh
12	South Africa Wind Energy	2016/01/01	94 871 MWh
13	West Coast 1 Wind Farm In South Africa	2015/04/01	293 400 MWh
14	Renewable Energy Carbon Programme For Africa (RECPA)	2015/03/01	140 180 MWh
15	Grid Connected Photovoltaic (PV) Renewable Electricity Generating Facilities Poa	2015/01/09	20 500 MWh
16	South African Large Scale Grid Connected Solar Park Programme	2015/01/01	66 395 MWh
17	Cookhouse Wind Farm In South Africa	2014/11/19	372 500 MWh
18	Kathu Grid Connected 100 MW Solar Park, South Africa	2014/11/01	260 074 MWh
19	Red Cap Kouga Wind Farm	2014/04/06	290 500 MWh
20	Karoo Renewable Energy Facility (Nobelsfontein Solar PV)	2014/01/01	94 527 MWh
21	Karoo Renewable Energy Facility (Nobelsfontein Wind)	2014/01/01	878 438 MWh
22	Dassieklip Wind Energy Facility In South Africa	2014/01/01	74 267 MWh
23	South African Wind Power Projects	2014/01/01	93 093 MWh
24	Prieska Grid Connected 20 MW Solar Park, South Africa	2013/11/01	39 483 MWh
25	Bokpoort	2013/10/01	224 000 MWh
26	Hopefield Wind Energy Facility In South Africa	2013/10/01	188 500 MWh
27	Southern African Renewable Energy (SARE) Programme	2013/07/12	54 785 MWh
28	Wind And Solar Poa In South Africa	2013/07/01	18 000 MWh
29	Coega IDZ Windfarm	2010/06/01	5 500 MWh
	TOTAL		41 768 562 MWh

As the annual electricity generation of this set comprises 20.6% of the annual electricity generation of the project electricity system (i.e. $AEG_{SET-sample-CDM} > 0.2 \times AEG_{total}$), this sample group is used to calculate the build margin.

The build margin emission factor was calculated as the generation-weighted average emission factor (tCO_2/MWh) for power units m during 2024, which is the most recent year y for which electricity generation data is available, using Equation 6:

$$EF_{grid,BM,y} = \frac{\sum m EG_{m,y} \times EF_{EL,m,y}}{\sum m EG_{m,y}}$$

Equation 6 (Tool 07 77 Equation 15)

Where:

$EF_{grid,BM,y}$	=	Build margin CO_2 emission factor in year y (tCO_2/MWh)
$EG_{m,y}$	=	Net quantity of electricity delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (tCO_2/MWh)
m	=	Power units included in the build margin
y	=	Most recent historical year for which electricity generation data is available

The results obtained are: $EF_{grid,BM,y} = 0.78$

$EG_{m,y}$ and $EF_{EL,m,y}$ were used as shown in Table 10.

5.1.6 Step 6: Calculate the combined margin (CM) emission factor

The combined margin (CM) emission factor ($EF_{grid,CM,y}$) was calculated as the weighted average CM, as per Equation 7.

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM}$$

Equation 7 (Tool 07 85 Equation 16)

As per VCSVT011, the following weighted values (for solar power generation) were used for w_{OM} and w_{BM} :

- $w_{OM} = 0.5$
- $w_{BM} = 0.5$

The results obtained are: $EF_{grid,CM,y} = 0.96 tCO_2/MWh$

5.2 Project Emissions

As per the approved consolidated Methodology ACM0002, “Grid-connected electricity generation from renewable sources” (Version 22.0) para 40: “For most renewable energy power generation project activities, $PE_y = 0$. However, some project activities may involve project emissions that can be significant. These emissions shall be accounted for as project emissions by using the following equation:

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y} + PE_{BESS,y} + PE_{PSP,y}$$

Where,

PE_y = Project emissions in year y (t CO₂e/yr)

$PE_{FF,y}$ = Project emissions from fossil fuel consumption in year y (tCO₂/yr)

$PE_{GP,y}$ = Project emissions from the operation of dry, flash steam or binary geothermal power plants in year y (tCO₂e/yr)

$PE_{HP,y}$ = Project emissions from water reservoirs of hydro power plants in year y (tCO₂e/yr)

$PE_{BESS,y}$ = Project emissions from charging of a BESS using electricity from the grid or from fossil fuel electricity generators (tCO₂e/yr)

$PE_{PSP,y}$ = Project emissions from utilizing electricity from the grid for pumping operation of PSP in excess to the production of the renewable power plant operating in coordination with the PSP (t CO₂e/yr)

As the project activity is the installation of a new grid-connected solar Power plant and does not involve any project emissions from fossil fuel, operation of dry, flash steam or binary geothermal power plants, and from water reservoirs of hydro power plants. Therefore $PE_{FF,y}$, $PE_{GP,y}$, $PE_{HP,y}$, $PE_{BESS,y}$, $PE_{PSP,y}$ are equal to zero and thus, $PE_y = 0$.

5.3 Leakage Emissions

As per the applied methodology, no other leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g.: extraction, processing, transport etc.) are neglected.

5.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

As per the paragraph 72 of the methodology ACM0002 “Grid-connected electricity generation from renewable sources” Version 22.0 emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y \text{ (Equation (17))}$$

Where:

ER_y = Emission reductions in year y (t CO₂e/yr)

BE_y = Baseline emissions in year y (t CO₂e/yr)

PE_y = Project emissions in year y (t CO₂e/yr)

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
18-NOV-2023 to 31-Dec-2023	83 913	0	0	83 913	0	83 913
01-Jan-2024 to 31-Dec-2024	824 868	0	0	824 868	0	824 868
01-Jan-2025 to 28-FEB-2025	136 418	0	0	136 418	0	136 418
Total	1 045 198	0	0	1 045 198	0	1 045 198

6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	$EF_{grid,BM,y}$
Data unit	tCO ₂ /MWh
Description	Build margin CO ₂ emission factor in year y
Source of data	Calculated using Equation 6, as per TOOL07 “Tool to calculate the emission factor for an electricity system”
Value applied:	0.78
Justification of choice of data or description of measurement methods and procedures applied	The BM was calculated as per option 1 (paragraph 72) of TOOL07 “Tool to calculate the emission factor for an electricity system”. An ex-ante approach was followed, based on the most recent information available on units already built. The BM will be updated for the second crediting period based on the most recent information available at that time. Capacity additions from retrofits of power plants are not included in the calculation of the build margin emission factor.
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO ₂ /MWh
Description	Operating margin CO ₂ emission factor in year y
Source of data	Calculated using Equation 1, as per TOOL07 “Tool to calculate the emission factor for an electricity system” Data Sources: Atmospheric Emission License (EAL) reports – Eskom Data Portal

Value applied:	1.11
Justification of choice of data or description of measurement methods and procedures applied	The simple OM emission factor was calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units. This means that only Eskom's coal-based power plants were included in the calculation. The OM was calculated as per option A (paragraph 47.a) of TOOL07 "Tool to calculate the emission factor for an electricity system", based on the net electricity generation and a CO₂ emission factor of each power unit.
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ /MWh
Description	Combine margin CO ₂ emission factor in year y
Source of data	Calculated using Equation 7, as per TOOL07 "Tool to calculate the emission factor for an electricity system"
Value applied:	0.96
Justification of choice of data or description of measurement methods and procedures applied	The combined margin (CM) emission factor (EF_{grid,CM,y}) was calculated based on the weighted average CM method (paragraph 81.a) since the data to determine the BM is available. This was done as per equation 16 of TOOL07 "Tool to calculate the emission factor for an electricity system". Where WOM is equal to 0.75 and WBM is equal to 0.25, for the first crediting period and for subsequent crediting periods
Purpose of data	Calculation of baseline emissions

Comments	No additional comments
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Data / Parameter	Share _{LCMR}								
Data unit	Percent (%)								
Description	Share of the low cost/must run resources								
Source of data	Determined from electricity generation data reported in Eskom’s annual Integrated Report for 2023 [13] and records published on Eskom’s website [23] for each power station.[13] and records published on Eskom’s website [23] for each power station.								
Value applied	Annual LCMR (%) <table><tr><td>2022</td><td>2023</td><td>2024</td></tr><tr><td>21.3%</td><td>23.1%</td><td>23.6%</td></tr></table>			2022	2023	2024	21.3%	23.1%	23.6%
2022	2023	2024							
21.3%	23.1%	23.6%							
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa’s national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table above								
Purpose of data	To calculate baseline emissions, using Equation 3.								
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in TOOL 07.								

Data / Parameter	EG _{LCMR,y}
Data unit	MWh
Description	Electricity generation supplied to the project electricity system by the low cost/must run resources in year y
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and records published on Eskom's website [23] for each power station. Determined from electricity generation data reported in Eskom's annual

	Integrated Report for 2023 [13] and records published on Eskom's website [23] for each power station.		
Value applied	<i>EG_{LCMRy}</i> : 2024	49 751 000	
	<i>EG_{LCMRy}</i> : 2023	49 691 000	
	<i>EG_{LCMRy}</i> : 2022	48 092 000	
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table above		
Purpose of data	To calculate baseline emissions, using Equation 3.		
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in TOOL 07.		

Data / Parameter	total _y		
Data unit	MWh		
Description	Total electricity generation supplied to the project electricity system in year y		
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 and monthly records published on Eskom's website [10] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and records published on Eskom's website [23] for each power station Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and records published on Eskom's website [23] for each power station		
Value applied	<i>total_y</i> : 2024	210 648	GWh/yr
	<i>total_y</i> : 2023	215 318	GWh/yr
	<i>total_y</i> : 2022	226 226	GWh/yr
Justification of choice of data or description of	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from		

measurement methods and procedures applied	Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table above
Purpose of data	To calculate baseline emissions, using Equation 3.
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in TOOL 07.

Data / Parameter	Y
Data unit	Year
Description	The most recent year for which data is available
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [24] and monthly records published on Eskom's website for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [24] and records published on Eskom's website [23] for each power station Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [24] and monthly records published on Eskom's website for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [24] and records published on Eskom's website [23] for each power station
Value applied	2024
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table above
Purpose of data	To calculate baseline emissions, using Equation 3.
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in TOOL 07.

Data / Parameter	$EG_{m,y}$			
Data unit	MWh			
Description	Net electricity delivered to the grid by power unit m in year y			
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [24] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [24] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements.			
Value applied	As per Table			
	Plant Name	Electricity generated (MWh/yr) 2022	Electricity generated (MWh/yr) 2023	Electricity generated (MWh/yr) 2024
	Arnot	9 085 064	7 530 158	7 530 158
	Duvha	9 822 489	9 555 266	9 555 266
	Hendrina	4 738 179	2 136 620	2 136 620
	Kendal	15 139 746	15 750 630	15 750 630
	Kriel	12 597 519	10 727 877	10 727 877
	Lethabo	22 890 481	21 367 191	21 367 191
	Matimba	23 180 336	23 257 630	23 257 630
	Majuba	19 823 301	15 540 628	15 540 628
	Matla	18 412 664	15 170 449	15 170 449
	Tutuka	8 524 715	7 564 337	7 564 337
	Camden	6 670 341	7 378 745	7 378 745
	Grootvlei	2 445 162	2 416 293	2 416 293
	Komati	608 728	369 330	369 330
	Kusile	7 992 189	9 119 476	9 119 476
	Medupi	22 637 675	23 246 212	23 246 212
	Acacia	873	1 069	1 069

	Port Rex	937	3 010	3 010
	Ankerlig	953 788	1 332 491	1 332 491
	Gourikwa	870 396	1 681 217	1 681 217
	Total	186 394 583	171 134 921	174 148 629
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table			
Purpose of data	To calculate baseline emissions, using Equation 4.			
Comments	The generation-weighted average CO ₂ emissions per unit net electricity generation (t CO ₂ /MWh) of all generating power plants serving the system, not including low-cost/must-run power plants is used to determine the simple OM emission factor, using Equation 3 under Option A from section 48 in TOOL 07.			

Data / Parameter	$EF_{EL,m,y}$
Data unit	tCO ₂ /MWh
Description	CO ₂ emission factor of power unit m in year y
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [24] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements.

Value applied	As per Table 8			
	Plant Name	Emission factor 2023 (tCO ₂ /MWh)	Emission factor 2022 (tCO ₂ /MWh)	Emission factor 2021 (tCO ₂ /MWh)
	Arnot	1.17	1.16	1.14
	Duvha	1.11	1.13	1.13
	Hendrina	1.35	1.24	1.20
	Kendal	1.02	1.05	1.04
	Kriel	1.32	1.28	1.22
	Lethabo	1.00	0.97	0.95
	Matimba	1.02	1.04	0.99
	Majuba	1.19	1.13	1.09
	Matla	1.30	1.27	0.99
	Tutuka	1.18	1.23	1.22
	Camden	1.10	1.16	1.25
	Grootvlei	1.21	1.27	1.24
	Komati	1.01	1.03	1.06
	Kusile	1.01	1.30	1.09
	Medupi	0.88	0.92	0.91
	Acacia	1.17	1.16	1.14
	Port Rex	1.17	1.16	1.14
	Ankerlig	1.17	1.16	1.14
	Gourikwa	1.17	1.16	1.14
Justification of choice of data or description of	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power			

measurement methods and procedures applied	Producers (IPPs) and for selling electricity to users. The power plants m are shown in Table above
Purpose of data	To calculate baseline emissions, using Equation 5.
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in TOOL 07 is applied.

Data / Parameter	$FC_{i,m,y}$
Data unit	Tonnes
Description	Amount of fuel type i consumed by power unit m in year y
Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and monthly and annual records published on Eskom's website [25] for each power station to fulfil the Atmospheric emission license (EAL) requirements Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and monthly and annual records published on Eskom's website [25] for each power station to fulfil the Atmospheric emission license (EAL) requirements Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [24] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [24] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements.

Value applied	As per Table 9						
	Plant Name	Coal usage (t) 2022	Coal usage (t) 2023	Coal usage (t) 2024	Fuel oil usage (t) 2022	Fuel oil usage (t) 2023	Fuel oil usage (t) 2024
	Arnot	5 186 766	4 293 316	4 828 094	37 792.2	45 179.7	46 635
	Duvha	5 732 195	5 428 210	6 802 842	39 550.9	71 179.4	72 479
	Hendrina	3 024 981	1 486 626	1 426 720	11 012.5	7 722.1	9 262
	Kendal	9 362 475	9 395 998	7 712 146	61 174.9	106 402.9	113 212
	Kriel	7 838 445	6 865 127	5 183 313	39 516.2	53 949.1	56 540
	Lethabo	16 277 939	15 646 686	16 048 047	12 792.0	14 075.3	13 252
	Matimba	13 430 751	13 179 578	10 674 314	9 944.3	14 066.5	18 715
	Majuba	11 924 693	9 831 151	11 304 146	149 002.7	140 169.2	19 808
	Matla	10 842 908	9 536 046	10 147 328	14 520.0	15 923.0	19 578
	Tutuka	5 510 214	4 697 013	3 879 070	68 779.3	64 592.8	62 137
	Camden	3 932 224	4 129 344	4 402 710	57 014.4	49 016.8	56 731
	Grootvlei	1 544 269	1 518 602	1 064 085	61 318.1	21 199.1	12 594
	Komati	335 833	197 954	-	1 332.0	1 773.0	-
	Kusile	4 783 403	5 035 930	5 481 821	534 043.2	10 351.0	26 640
	Medupi	11 351 788	11 181 866	11 065 548	22 637.7	23 246.2	20 489
	Plant Name	Diesel usage (l) 2022		Diesel usage (l) 2023		Diesel usage (l) 2024	
	Acacia	350 693		414 371		936 177	
	Port Rex	364 144		1 093 406		1 458 236	
	Ankerlig	300 686 482		418 610 227		507 827 227	
	Gourikwa	269 555 564		515 526 621		622 893 869	
	Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa’s national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The power plants m are shown in Table above					
Purpose of data	To calculate baseline emissions, using Equation 5.						
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in TOOL 07 is applied.						

Data / Parameter	NCV _{i,y}
Data unit	GJ/tonne

Description	Net calorific value of fuel type i in year y								
Source of data	IPCC default factors for coal, heavy fuel oil and diesel , [26] IPCC default factors for coal, heavy fuel oil and diesel , [26]								
Value applied	For coal as per plant in Table <table border="1"> <tr> <td>Acacia</td><td>35.5</td></tr> <tr> <td>Port Rex</td><td>35.5</td></tr> <tr> <td>Ankerlig</td><td>35.5</td></tr> <tr> <td>Gourikwa</td><td>35.5</td></tr> </table>	Acacia	35.5	Port Rex	35.5	Ankerlig	35.5	Gourikwa	35.5
Acacia	35.5								
Port Rex	35.5								
Ankerlig	35.5								
Gourikwa	35.5								
Justification of choice of data or description of measurement methods and procedures applied	Tier 2 NCV (for South Africa) only available for coal. Tier 1 used for other NCVs								
Purpose of data	To calculate baseline emissions, using Equation 5.								
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in TOOL 07 is applied.								

Data / Parameter	$AE G_{total}$
Data unit	GWh
Description	Total electricity generation of all power plants directly supplying the system in year y
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [24] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements.
Value applied	174,148
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table above.
Purpose of data	To calculate baseline emissions
Comments	No additional comments

Data / Parameter	SET_5 units												
Data unit	Dimensionless												
Description	In line with VT0011, the set of five power units, including power units registered as VCS or other GHG programs, that started to supply electricity to the grid most recently was identified												
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [24] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. [13] Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [13] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 and monthly and annual records published on Eskom's website [23] for each power station to fulfil the Atmospheric emission license (EAL)												
Value applied	<table> <tr> <th>Plant name</th><th>Commissioning Date</th></tr> <tr> <td>TWE Golden Valley Wind Power Project</td><td>2021/05/01</td></tr> <tr> <td>GS11949 Small-scale solar electrical programme, South Africa</td><td>2020/12/16</td></tr> <tr> <td>Solar Energy Programme for South Africa</td><td>2019/02/01</td></tr> <tr> <td>Longyuan Mulilo De Aar Maanhaarberg Wind Energy Facility</td><td>2017/10/31</td></tr> <tr> <td>Longyuan Mulilo De Aar 2 North Wind Energy Facility</td><td>2017/10/31</td></tr> </table>	Plant name	Commissioning Date	TWE Golden Valley Wind Power Project	2021/05/01	GS11949 Small-scale solar electrical programme, South Africa	2020/12/16	Solar Energy Programme for South Africa	2019/02/01	Longyuan Mulilo De Aar Maanhaarberg Wind Energy Facility	2017/10/31	Longyuan Mulilo De Aar 2 North Wind Energy Facility	2017/10/31
Plant name	Commissioning Date												
TWE Golden Valley Wind Power Project	2021/05/01												
GS11949 Small-scale solar electrical programme, South Africa	2020/12/16												
Solar Energy Programme for South Africa	2019/02/01												
Longyuan Mulilo De Aar Maanhaarberg Wind Energy Facility	2017/10/31												
Longyuan Mulilo De Aar 2 North Wind Energy Facility	2017/10/31												
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in												
Purpose of data	To calculate baseline emissions												

Comments	This set of power units generate 19.2% of AEG_{total} in 2024.
Data / Parameter	$AEG_{set-5-units}$
Data unit	GWh
Description	Annual electricity generation of the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [24] and monthly records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements.
Value applied	1,643
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table above
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 19,2% of AEG_{total} in 2023.

Data / Parameter	$SET_{\geq 20\%}$		
Data unit	Dimensionless		
Description	The set of power units, including power units registered as CDM/VCS project activities, and that comprise 20% of AEG_{total}		
Source of data	Determined from IPP database [27].		
Value applied	No.	Plant name	Commissioning Date
			Power Produced in Year: 2024
	Eskom Fleet Plants		
	1	Gourikwa	2007/07/01
	2	Medupi	2015/08/22
	3	Kusile	2017/08/30
			2 038 316 MWh
			22 626 444 MWh
			10 262 515 MWh

IPP Power Plants			
4	Green Power For South Africa	2013/05/31	1 219 826 MWh
5	South African Grid Connected Wind Farm Programme	2014/04/01	318 056 MWh
6	TWE Golden Valley Wind Power Project	2021/05/01	509 303 MWh
7	GS11949 Small-Scale Solar Electrical Programme, South Africa	2020/12/16	22 500 MWh
8	Solar Energy Programme For South Africa	2019/02/01	381 930 MWh
9	Longyuan Mulilo De Aar Maanhaarberg Wind Energy Facility	2017/10/31	290 500 MWh
10	Longyuan Mulilo De Aar 2 North Wind Energy Facility	2017/10/31	439 600 MWh
11	Amakhala Emoyeni Grid Connected 138.6 MW Wind Farm, Phase 1, South Africa	2016/06/15	450 559 MWh
12	South Africa Wind Energy	2016/01/01	94 871 MWh
13	West Coast 1 Wind Farm In South Africa	2015/04/01	293 400 MWh
14	Renewable Energy Carbon Programme For Africa (RECPA)	2015/03/01	140 180 MWh
15	Grid Connected Photovoltaic (PV) Renewable Electricity Generating Facilities Poa	2015/01/09	20 500 MWh
16	South African Large Scale Grid Connected Solar Park Programme	2015/01/01	66 395 MWh
17	Cookhouse Wind Farm In South Africa	2014/11/19	372 500 MWh
18	Kathu Grid Connected 100 MW Solar Park, South Africa	2014/11/01	260 074 MWh
19	Red Cap Kouga Wind Farm	2014/04/06	290 500 MWh
20	Karoo Renewable Energy Facility (Nobelsfontein Solar PV)	2014/01/01	94 527 MWh
21	Karoo Renewable Energy Facility (Nobelsfontein Wind)	2014/01/01	878 438 MWh
22	Dassieklip Wind Energy Facility In South Africa	2014/01/01	74 267 MWh
23	South African Wind Power Projects	2014/01/01	93 093 MWh
24	Prieska Grid Connected 20 MW Solar Park, South Africa	2013/11/01	39 483 MWh
25	Bokpoort	2013/10/01	224 000 MWh
26	Hopefield Wind Energy Facility In South Africa	2013/10/01	188 500 MWh
27	Southern African Renewable Energy (SARE) Programme	2013/07/12	54 785 MWh
28	Wind And Solar Poa In South Africa	2013/07/01	18 000 MWh
29	Coega IDZ Windfarm	2010/06/01	5 500 MWh
	TOTAL		41 768 562 MWh

Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 12
Purpose of data	To calculate baseline emissions
Comments	This set of power units generate 20.6% of AEG_{total} in 2024.

Data / Parameter	$AEG_{set \geq 20\%}$
Data unit	GWh
Description	Annual electricity generation of the set of power units, including power units registered as CDM project activities, and excluding those commissioned more than 10 years ago, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [24] and IPP database [27]
Value applied	41,768,562
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 12
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 20.6% of AEG_{total} in 2024.

Data / Parameter	SET_{sample}
Data unit	Dimensionless
Description	The set of power units, (between $SET_{5 \text{ units}}$ and $SET_{\geq 20\%}$) that comprises the larger annual electricity generation.
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 and monthly records

	published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements.																																																																																																
Value applied	<table><tr><th colspan="3">Eskom Fleet Plants</th></tr><tr><td>1</td><td>Gourikwa</td><td>2007/07/01</td></tr><tr><td>2</td><td>Medupi</td><td>2015/08/22</td></tr><tr><td>3</td><td>Kusile</td><td>2017/08/30</td></tr><tr><th colspan="3">IPP Power Plants</th></tr><tr><td>4</td><td>Green Power For South Africa</td><td>2013/05/31</td></tr><tr><td>5</td><td>South African Grid Connected Wind Farm Programme</td><td>2014/04/01</td></tr><tr><td>6</td><td>TWE Golden Valley Wind Power Project</td><td>2021/05/01</td></tr><tr><td>7</td><td>GS11949 Small-Scale Solar Electrical Programme, South Africa</td><td>2020/12/16</td></tr><tr><td>8</td><td>Solar Energy Programme For South Africa</td><td>2019/02/01</td></tr><tr><td>9</td><td>Longyuan Mulilo De Aar Maanhaarberg Wind Energy Facility</td><td>2017/10/31</td></tr><tr><td>10</td><td>Longyuan Mulilo De Aar 2 North Wind Energy Facility</td><td>2017/10/31</td></tr><tr><td>11</td><td>Amakhala Emoyeni Grid Connected 138.6 MW Wind Farm, Phase 1, South Africa</td><td>2016/06/15</td></tr><tr><td>12</td><td>South Africa Wind Energy</td><td>2016/01/01</td></tr><tr><td>13</td><td>West Coast 1 Wind Farm In South Africa</td><td>2015/04/01</td></tr><tr><td>14</td><td>Renewable Energy Carbon Programme For Africa (RECPA)</td><td>2015/03/01</td></tr><tr><td>15</td><td>Grid Connected Photovoltaic (PV) Renewable Electricity Generating Facilities Poa</td><td>2015/01/09</td></tr><tr><td>16</td><td>South African Large Scale Grid Connected Solar Park Programme</td><td>2015/01/01</td></tr><tr><td>17</td><td>Cookhouse Wind Farm In South Africa</td><td>2014/11/19</td></tr><tr><td>18</td><td>Kathu Grid Connected 100 MW Solar Park, South Africa</td><td>2014/11/01</td></tr><tr><td>19</td><td>Red Cap Kouga Wind Farm</td><td>2014/04/06</td></tr><tr><td>20</td><td>Karoo Renewable Energy Facility (Nobelsfontein Solar PV)</td><td>2014/01/01</td></tr><tr><td>21</td><td>Karoo Renewable Energy Facility (Nobelsfontein Wind)</td><td>2014/01/01</td></tr><tr><td>22</td><td>Dasseldorp Wind Energy Facility In South Africa</td><td>2014/01/01</td></tr><tr><td>23</td><td>South African Wind Power Projects</td><td>2014/01/01</td></tr><tr><td>24</td><td>Prieska Grid Connected 20 MW Solar Park, South Africa</td><td>2013/11/01</td></tr><tr><td>25</td><td>Bokpoort</td><td>2013/10/01</td></tr><tr><td>26</td><td>Hopefield Wind Energy Facility In South Africa</td><td>2013/10/01</td></tr><tr><td>27</td><td>Southern African Renewable Energy (SARE) Programme</td><td>2013/07/12</td></tr><tr><td>28</td><td>Wind And Solar Poa In South Africa</td><td>2013/07/01</td></tr><tr><td>29</td><td>Coega IDZ Windfarm</td><td>2010/06/01</td></tr><tr><td colspan="2">TOTAL</td><td></td></tr></table>	Eskom Fleet Plants			1	Gourikwa	2007/07/01	2	Medupi	2015/08/22	3	Kusile	2017/08/30	IPP Power Plants			4	Green Power For South Africa	2013/05/31	5	South African Grid Connected Wind Farm Programme	2014/04/01	6	TWE Golden Valley Wind Power Project	2021/05/01	7	GS11949 Small-Scale Solar Electrical Programme, South Africa	2020/12/16	8	Solar Energy Programme For South Africa	2019/02/01	9	Longyuan Mulilo De Aar Maanhaarberg Wind Energy Facility	2017/10/31	10	Longyuan Mulilo De Aar 2 North Wind Energy Facility	2017/10/31	11	Amakhala Emoyeni Grid Connected 138.6 MW Wind Farm, Phase 1, South Africa	2016/06/15	12	South Africa Wind Energy	2016/01/01	13	West Coast 1 Wind Farm In South Africa	2015/04/01	14	Renewable Energy Carbon Programme For Africa (RECPA)	2015/03/01	15	Grid Connected Photovoltaic (PV) Renewable Electricity Generating Facilities Poa	2015/01/09	16	South African Large Scale Grid Connected Solar Park Programme	2015/01/01	17	Cookhouse Wind Farm In South Africa	2014/11/19	18	Kathu Grid Connected 100 MW Solar Park, South Africa	2014/11/01	19	Red Cap Kouga Wind Farm	2014/04/06	20	Karoo Renewable Energy Facility (Nobelsfontein Solar PV)	2014/01/01	21	Karoo Renewable Energy Facility (Nobelsfontein Wind)	2014/01/01	22	Dasseldorp Wind Energy Facility In South Africa	2014/01/01	23	South African Wind Power Projects	2014/01/01	24	Prieska Grid Connected 20 MW Solar Park, South Africa	2013/11/01	25	Bokpoort	2013/10/01	26	Hopefield Wind Energy Facility In South Africa	2013/10/01	27	Southern African Renewable Energy (SARE) Programme	2013/07/12	28	Wind And Solar Poa In South Africa	2013/07/01	29	Coega IDZ Windfarm	2010/06/01	TOTAL		
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Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 12																																																																																																
Purpose of data	To calculate baseline emissions																																																																																																
Comments	This set of power units generated 41,768 GWh in 2024, comprising 20.6% of AEG _{total} in 2024.																																																																																																

Data / Parameter	$AEG_{set\ sample}$
Data unit	GWh
Description	Annual electricity generation of the set of power units, including power units registered as CDM project activities that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [24] and IPP database [27]
Value applied	41,768
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated >20% of AEG_{total} in 2024.

Data / Parameter	w_{BM}
Data unit	Percentage (%)
Description	Weighting of build margin emissions factor
Source of data	VT0011 Paragraph 86
Value applied	50%
Justification of choice of data or description of measurement methods and procedures applied	Default value provided in VT0011
Purpose of data	To calculate baseline emissions, using Equation 7
Comments	Owing to the intermittent and non-dispatchable nature of solar power generation, the weighting of the build margin emission factor is set at 50% for the first crediting period.

Data / Parameter	w_{OM}
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Data unit	Percentage (%)
Description	Weighting of operating margin emissions factor
Source of data	VT0011 Paragraph 86
Value applied	50%
Justification of choice of data or description of measurement methods and procedures applied	Default value provided in VT0011
Purpose of data	To calculate baseline emissions, using Equation 7
Comments	Owing to the intermittent and non-dispatchable nature of solar power generation, the weighting of the operating margin emission factor is set at 50% for the first crediting period.

Data / Parameter	BE_y
Data unit	tCO ₂ e/yr
Description	Emission reductions in year y
Source of data	Quantified using Equation 1

Value applied	<table> <tr> <th>Vintage period</th><th>Estimated baseline emissions (tCO₂e)</th></tr> <tr> <td>18-NOV-2023 to 31-Dec-2023</td><td>83 913</td></tr> <tr> <td>01-Jan-2024 to 31-Dec-2024</td><td>824 868</td></tr> <tr> <td>01-Jan-2025 to 28-FEB-2025</td><td>136 418</td></tr> <tr> <td>Total</td><td>1 045 198</td></tr> </table>	Vintage period	Estimated baseline emissions (tCO ₂ e)	18-NOV-2023 to 31-Dec-2023	83 913	01-Jan-2024 to 31-Dec-2024	824 868	01-Jan-2025 to 28-FEB-2025	136 418	Total	1 045 198
Vintage period	Estimated baseline emissions (tCO ₂ e)										
18-NOV-2023 to 31-Dec-2023	83 913										
01-Jan-2024 to 31-Dec-2024	824 868										
01-Jan-2025 to 28-FEB-2025	136 418										
Total	1 045 198										
Justification of choice of data or description of measurement methods and procedures applied	Equation provided in ACM0002.										
Purpose of data	To calculate baseline emissions, using Equation 1										
Comments	No additional comments										

6.2 Data and Parameters Monitored

Data / Parameter	$EG_{facility,y}$
Data unit	MWh/Year
Description	Quantity of net electricity generation supplied by the project to the grid in year y
Source of data	Electricity reading meters
Description of measurement methods	Measurement data is managed by the project's SCADA system. The measuring and monitoring equipment is supplied by a

and procedures to be applied	combination or OEM suppliers, including Sneider Electric, Schweitzer Engineering Laboratories (SEL), Landis+Gyr and ABB, and was installed by competent instrumentation Technicians and OEMs.
Frequency of monitoring/recording	Continuous monitoring and monthly reporting
Value applied	Ex-ante values for the first crediting period: 860, 000 MWh /yr
Monitoring equipment	The measuring and monitoring equipment is supplied by a combination or OEM suppliers, including Sneider Electric, Schweitzer Engineering Laboratories (SEL), Landis+Gyr and ABB, and was installed by competent instrumentation Technicians and OEMs
QA/QC procedures to be applied	Metered electricity production will be recorded by the plant on the DCS system. This will be checked against the Eskom meter at the independent switching station. Any discrepancies will immediately trigger fault finding and corrective action procedures. There is also a backup meter both on site and at the switching station which are used as an additional failsafe for QA/QC purposes. All reported electricity generation values will be checked by both Redstone personnel and Eskom personnel prior to billing and payment.
Purpose of data	Calculation of project emissions
Calculation method	Net electricity supplied to the grid by the project plant is directly measured and reported, not calculated.
Comments	No additional comments

Data / Parameter	PE_y
Data unit	tCO ₂ /yr
Description	Project emissions in year y
Source of data	Quantified using Equation 8

Description of measurement methods and procedures to be applied	Quantified, not measured
Frequency of monitoring/recording	Monthly
Value applied	0
Monitoring equipment	Quantified, not monitored
QA/QC procedures to be applied	No additional QA/QC would be applied to this equation
Purpose of data	Calculation of project emissions
Calculation method	Equation 8
Comments	For all renewable energy power generation project activities, emissions due to the use of fossil fuels for the backup generator can be neglected. This project does not involve geothermal power plants or water reservoirs of hydro power plants; therefore, project emissions for this renewable energy power generation will be zero.

Data / Parameter	ER_y
Data unit	tCO ₂ /yr
Description	Emission reductions in year y
Source of data	Quantified using Equation 9
Description of measurement methods and procedures to be applied	Quantified, not measured
Frequency of monitoring/recording	Monthly
Value applied	Ex-ante value is 860 000 per annum

Monitoring equipment	Quantified, not monitored
QA/QC procedures to be applied	No additional QA/QC would be applied to this equation
Purpose of data	Calculation of project emissions
Calculation method	Equation 9
Comments	The emission reductions determined at validation apply estimates of likely emissions reductions from the proposed project activity during the crediting period, using available ex-ante values. These will be revised during each verification using the actual data recorded (where applicable).

6.3 Monitoring Plan

Methods for measuring, recording, storing, aggregating, collating and reporting on monitored data and parameters.

The facility has a TGS Prediktor Supervisory Control and Data Acquisition (SCADA) system that ensures plant-wide integration of devices and allows for effective plant operations and monitoring and reporting. Apart from reporting for this project, the data will also be used for reconciliation and checking the billing from the national utility for electricity supplied to the grid. All data and reports will be stored and backed up in accordance with the SCADA system design, this will be done off site and is linked to a Global Operations Centre (GOC) hub remotely. The GOC monitors numerous projects that the PP operates globally. The location of the meters on site is shown below.

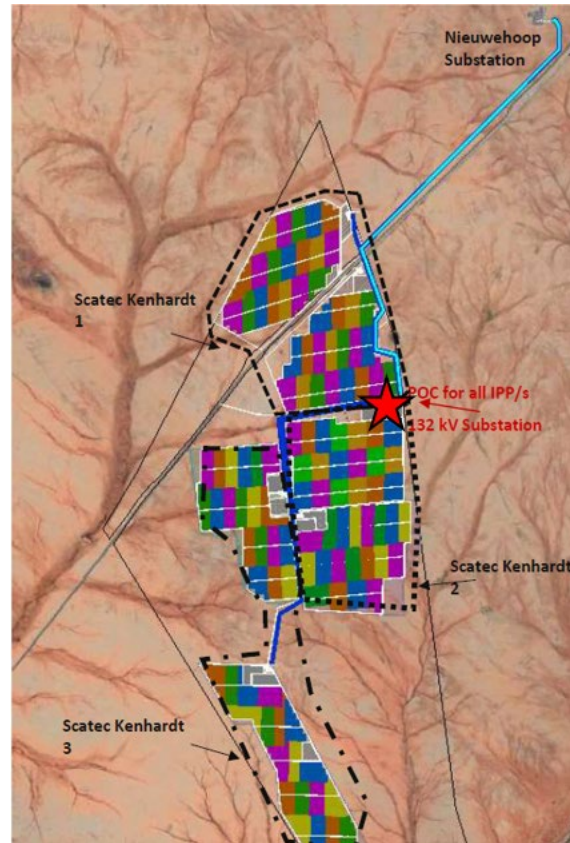


Figure 10: Meter locations on site (indicated by star shape)



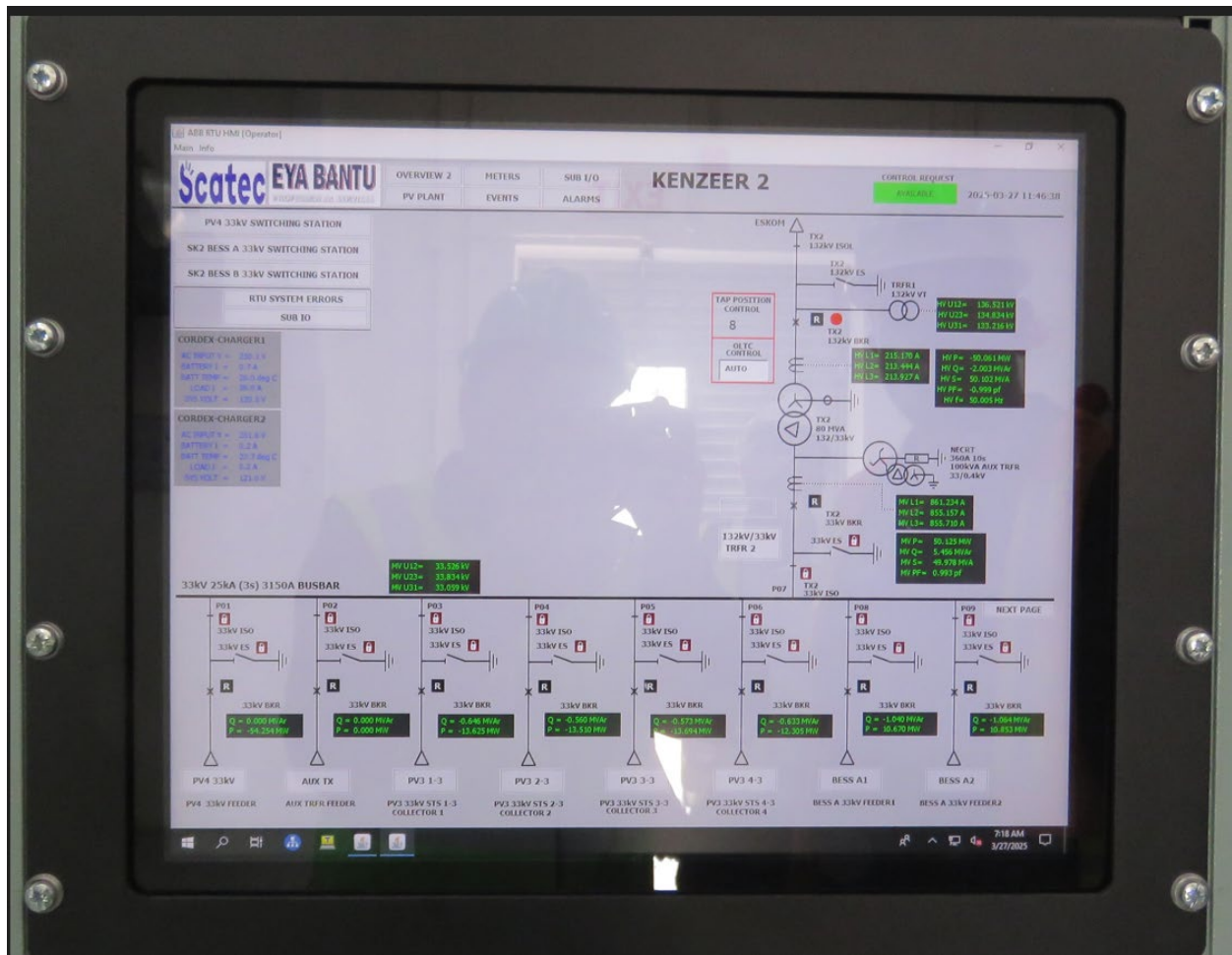


Figure 12: Indicative instrumentation SLD of metering configuration to 132kV

Measurement data is managed by the project's SCADA system. The measuring and monitoring equipment is supplied by a combination of OEM suppliers, including Sneider Electric, Schweitzer Engineering Laboratories (SEL), Landis+Gyr and ABB, and was installed by competent instrumentation Technicians and OEMs:

- Calibration under standard temperature and pressure conditions by OEM
- Installation and electrical connection by a certified external contractor
- Commissioning and final test under project conditions by the internal instrumentation team

All data is stored on a server on-site and also loaded onto a firewall protected cloud storage system. All measuring equipment strictly complies with their specified maintenance plans and calibration certificates are kept on record.

As per the applied methodology, electricity generation will be measured onsite at the outgoing switching station. This will be monitored on a continuous basis, with a monthly report being aggregated and

reported upon. All monitoring and reporting will be done at the GOC unit in the Western Cape. The check meters, owned by the national utility (Eskom), is located at the adjacent Eskom Rooidam 132 kV Switching Station within the perimeter fence which serves as the grid tie-in point. Meters at both locations have a back-up meter installed as a failsafe.

The electrical meters on site, calibrated in accordance with the manufacturer's specifications, will form part of the planned maintenance schedule for both checking and calibration. Instrument accuracy checks and verification will be conducted by Competent Instrumentation technicians and records kept on site. Calibrations will be conducted by competent technicians qualified under the national SANAS accreditation. The applicable national standard to be applied is SANS 474 [28]. This defines the required frequency of calibration in §4.7.4 for meters greater than 10MVA as being every 5 years. In addition to the SANS 474 national standard, the Power Purchase Agreement (PPA) regulates the calibration requirement under Section 12 – Metering. These requirements have been incorporated into the planned operations and maintenance schedules of the facility. While the PPA does not stipulate a calibration frequency, and places no obligation on the national utility, the project has opted to provide a calibration frequency of a maximum of every 5 years as per the SANS474 national standard. The calibrations are carried out by competent and SANAS accredited third party instrumentation companies and calibration certificates are issued and kept on record.

Organizational structure, responsibilities and competencies of the personnel that will be carrying out monitoring activities.

Data collection, consolidation and results analysis will be undertaken by a team who are adequately trained and competent. They will be responsible for the maintenance (planned and unplanned), checking and arranging calibration for plant equipment. They will also be responsible for maintaining records of calibrations and procurement of any instrumentation equipment that may be required for the operation of the plant. All instrument technicians will be suitably competent and records of their training and qualifications will be maintained by the Human Resources department. Only SANAS qualified organizations will conduct calibrations. These organizations will be part of an approved suppliers list and are also integrated into the certified ISO9001, ISO14001 and ISO45001 management systems.

Procedures for internal auditing and QA/QC

Metered electricity production will be recorded by the plant on the SCADA system. This will be checked against the Eskom meter at the independent Rooidam switching station. There is also a backup meter both on site and at the Rooidam switching station which are used as an additional failsafe for QA/QC purposes. All reported electricity generation values will be checked by both Scatec personnel and Eskom personnel prior to billing and payment. The final reporting checks are conducted off site at the Scatec GOC based in Cape Town.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Data / Parameter	$EG_{facility,y}$
Data unit	MWh/Year
Description	Quantity of net electricity generation supplied by the project to the grid in year y
Value applied:	1,088,748
Comments	From Plant reporting

Data / Parameter	ER_y
Data unit	tCO ₂ /yr
Description	Emission reductions in year y
Value applied:	1,045,198
Comments	Calculated from Equation 1

Data / Parameter	PE_y
Data unit	tCO ₂ /yr
Description	Emission reductions in year y
Value applied:	0

Comments	As per methodology
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Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO ₂ /MWh
Description	Combine margin CO ₂ emission factor in year y
Value applied:	0.96
Comments:	Calculated above

7.2 Baseline Emissions

The baseline emissions were quantified in accordance with the methodology ACM0002 version 22.0. For this project, it includes electricity delivered to the South African National grid. This was calculated as per the calculation steps described below.

Baseline Emission Calculation

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Equation 8 (ACM0002 Equation 11)

Where:

BE_y = Baseline emissions in year y (tCO₂/yr)

$EG_{PJ,y}$ = Net electricity fed into the grid supplied by the project activity in year y
(MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in

year y calculated using TOOL07 (t CO₂/MWh)

The results obtained are: 830 000 t CO₂/ annum, as shown below:

Parameter name	Value	Unit
BE_y	830, 000	t CO ₂ / yr
$EG_{PJ,y}$	860, 000	MWh/yr
$EF_{grid,CM,y}$	0.96	t CO ₂ /MWh

7.3 Project Emissions

As per the approved consolidated Methodology ACM0002, “Grid-connected electricity generation from renewable sources” (Version 22.0) para 40: “For most renewable energy power generation project activities, PE_y = 0. However, some project activities may involve project emissions that can be significant. These emissions shall be accounted for as project emissions by using the following equation:

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y} + PE_{BESS,y} + PE_{PSP,y}$$

Where,

PE_y = Project emissions in year y (t CO₂e/yr)

PE_{FF,y} = Project emissions from fossil fuel consumption in year y (tCO₂/yr)

PE_{GP,y} = Project emissions from the operation of dry, flash steam or binary geothermal power plants in year y (tCO₂e/yr)

PE_{HP,y} = Project emissions from water reservoirs of hydro power plants in year y (tCO₂e/yr)

PE_{BESS,y} = Project emissions from charging of a BESS using electricity from the grid or from fossil fuel electricity generators (tCO₂e/yr)

PE_{PSP,y} = Project emissions from utilizing electricity from the grid for pumping operation of PSP in excess to the production of the renewable power plant operating in coordination with the PSP (t CO₂e/yr)

As the project activity is the installation of a new grid-connected solar Power plant and does not involve any project emissions from fossil fuel, operation of dry, flash steam or binary geothermal power plants, and from water reservoirs of hydro power plants. Therefore PE_{FF,y}, PE_{GP,y}, PE_{HP,y}, PE_{BESS,y}, PE_{PSP,y} are equal to zero and thus, PE_y = 0.

7.4 Leakage Emissions

As per the applied methodology, no other leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g.: extraction, processing, transport etc.) are neglected.

7.5 GHG Emission Reductions and Carbon Dioxide Removals

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCU's (tCO ₂ e)	Removal VCU's (tCO ₂ e)	Total VCU's (tCO ₂ e)
18-NOV-2023 to 31-DEC-2023	83 913	0	0	83 913	0	83 913
01-Jan-2024 to 31-DEC-2024	824 868	0	0	824 868	0	824 868
01-Jan-2025 to 28-FEB-2025	136 418	0	0	136 418	0	136 418
Total	1 045 198	0	0	1 045 198	0	1 045 198

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Section	Information	Justification
	No Commercially sensitive information was included in this version of the PD	

APPENDIX 2: BIBLIOGRAPHY